EEE 324: Digital Signal Processing

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Week 4: Z-Transform

- Direct Z-Transform and ROC
- Z-Transform Properties
- Rational Z-Transform
- Inversion of Z-Transform
- Analysis of LTI systems in Z-domain

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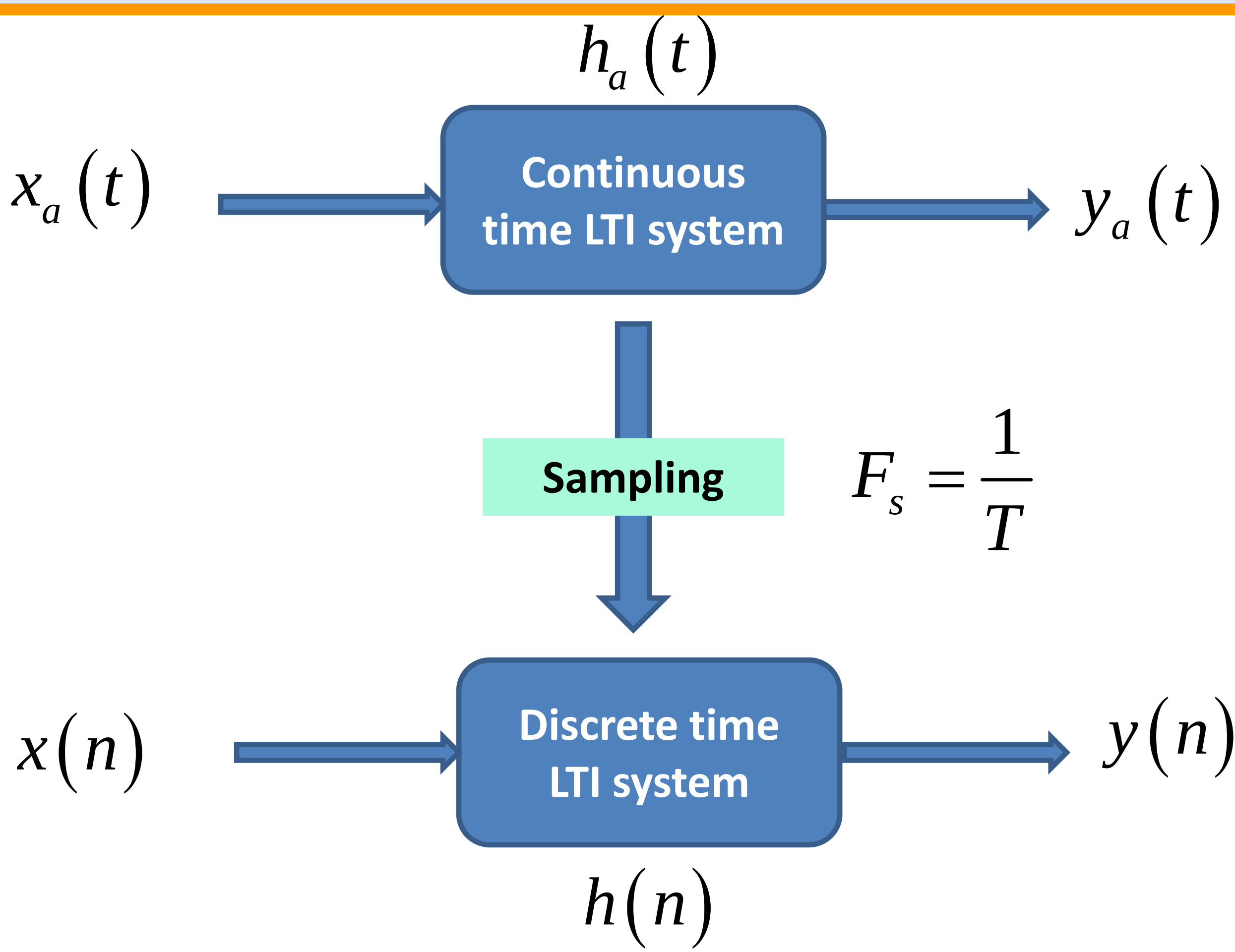
Z-Transform Introduction

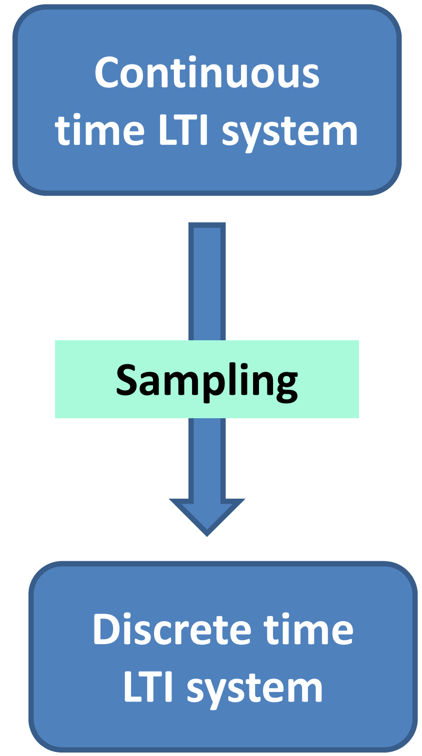


Z-Transform Introduction

Laplace Transform: $X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$

$$\boxed{z = e^{sT}}$$

Z Transform: $X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$



$$F_s = \frac{1}{T}$$

Z-Transform Introduction

Laplace Transform:

Essential mathematical tools

- Closed form mathematical description of system in transform domain.
- System Design
- System Analysis
  - System stability
  - Transient Response
  - Steady state behavior

Z Transform:

Continuous time LTI system

↓ **Sampling** $F_s = \frac{1}{T}$

Discrete time LTI system

Why Transforms ?

Why Transforms ?

Time Domain ⟷ **Frequency Domain**

Time-domain and frequency domain representation methods offer alternative insights into a system, and depending on the application it may be more convenient to use one method in preference to the other

- Time domain system analysis methods are based on differential and difference equations
- Frequency domain methods, mainly the Laplace transform, the Fourier transform, and the z-transform, describe a system in terms of its response to the individual frequency constituents of the input signal

Differential and difference equations are transformed into algebraic equations which are much easier to handle

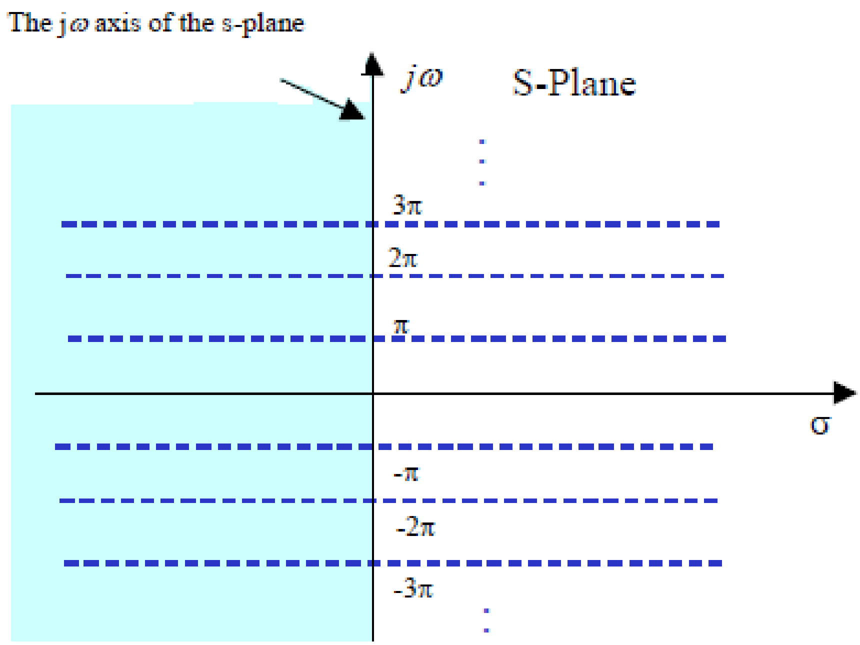
Differentiation and integration operations are performed through multiplication and division

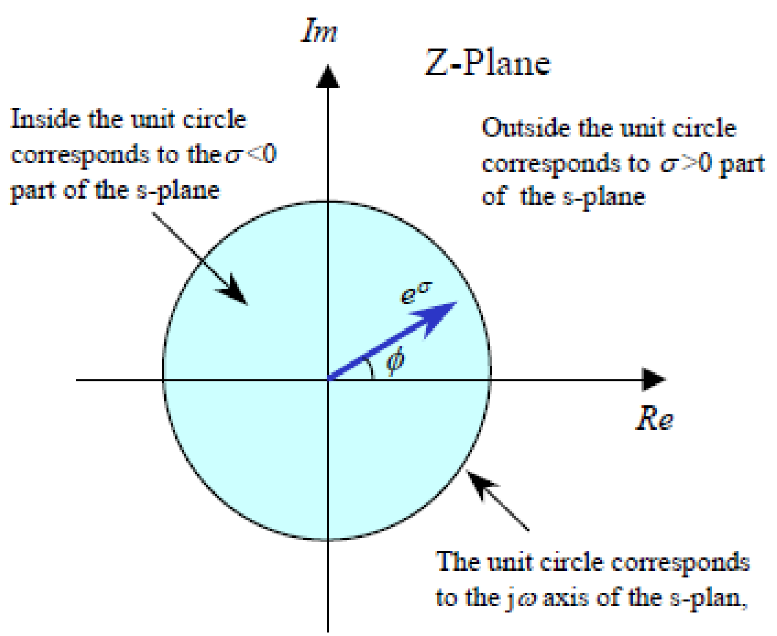
Z-Transform vs. Laplace Transform

$$z = e^{sT} = e^{T(\sigma + j\omega)}$$

Usually, we consider $T = 1$, so,

$$z = e^{\sigma} e^{j\omega} = re^{j\omega}$$



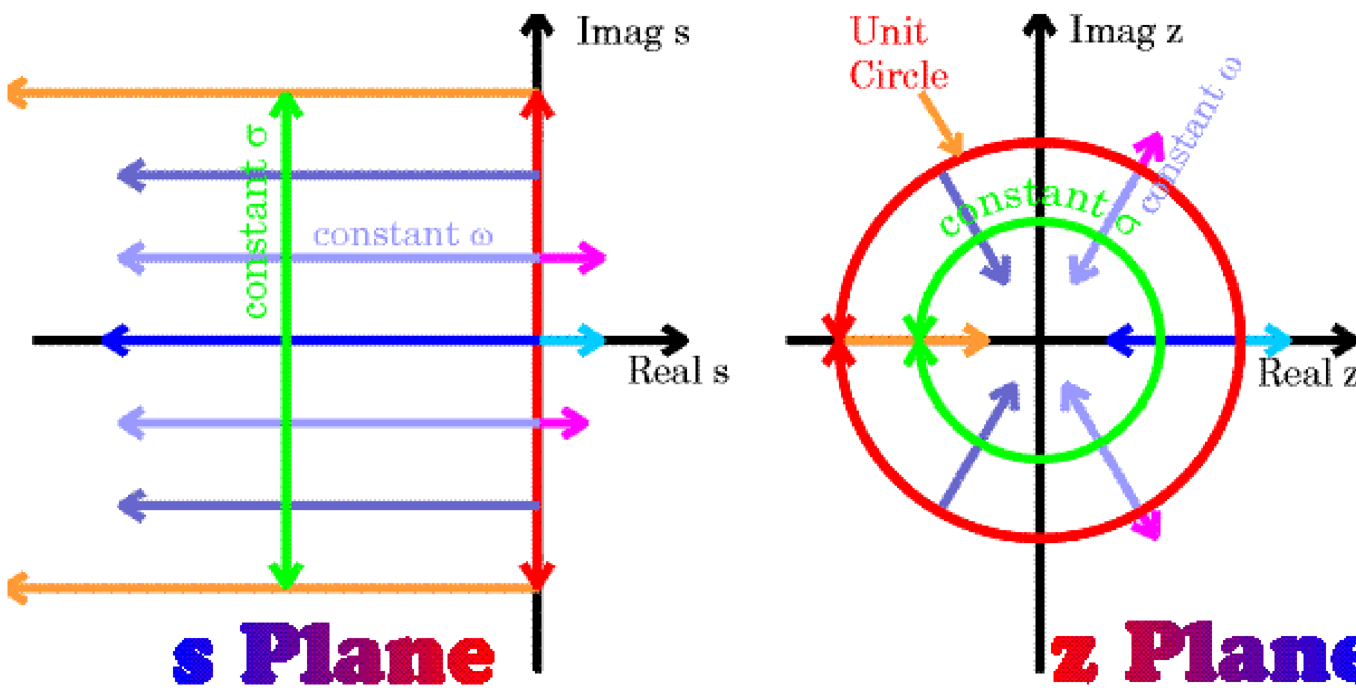


Z-Transform vs. Laplace Transform

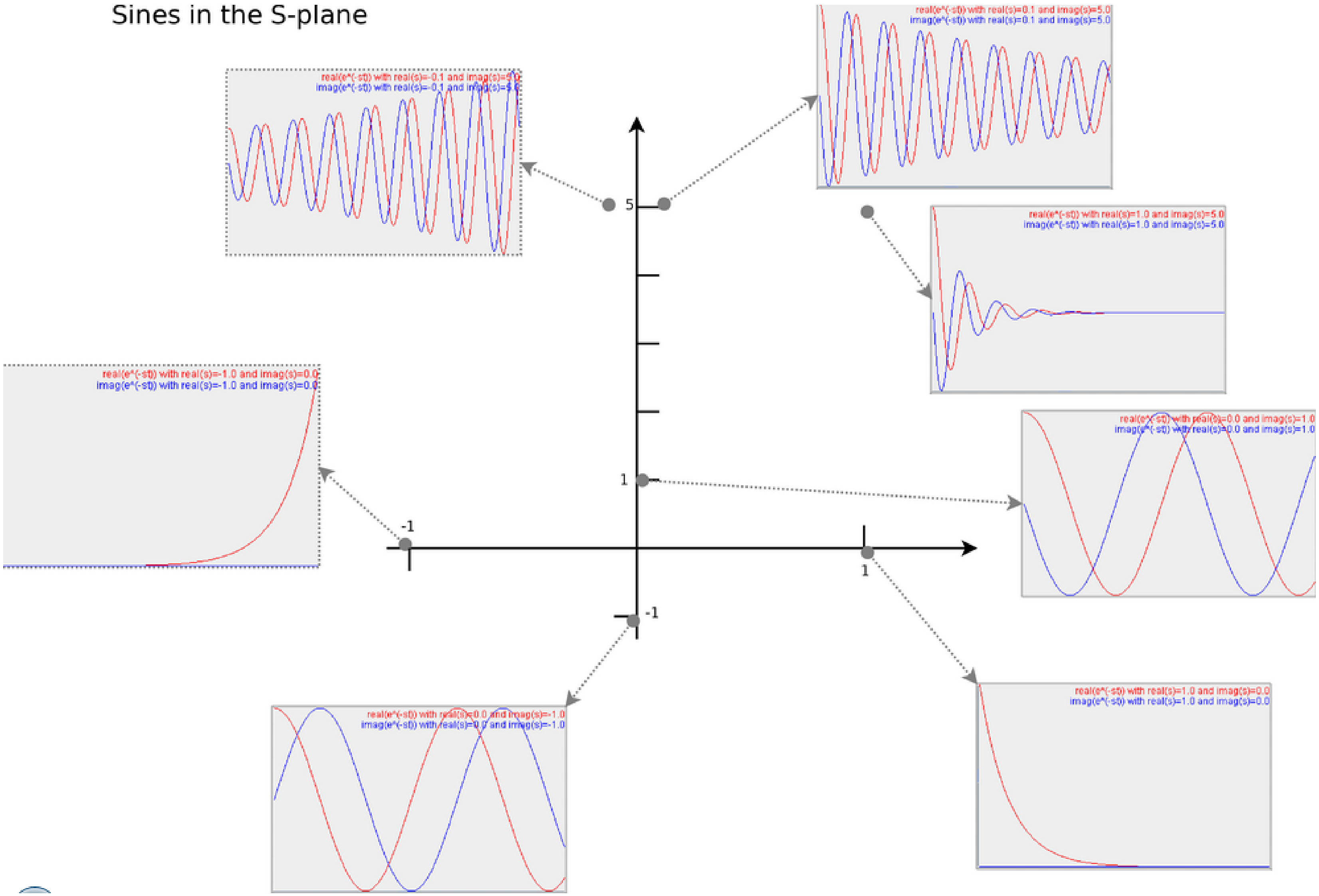
Both are complex planes

Cartesian Plane

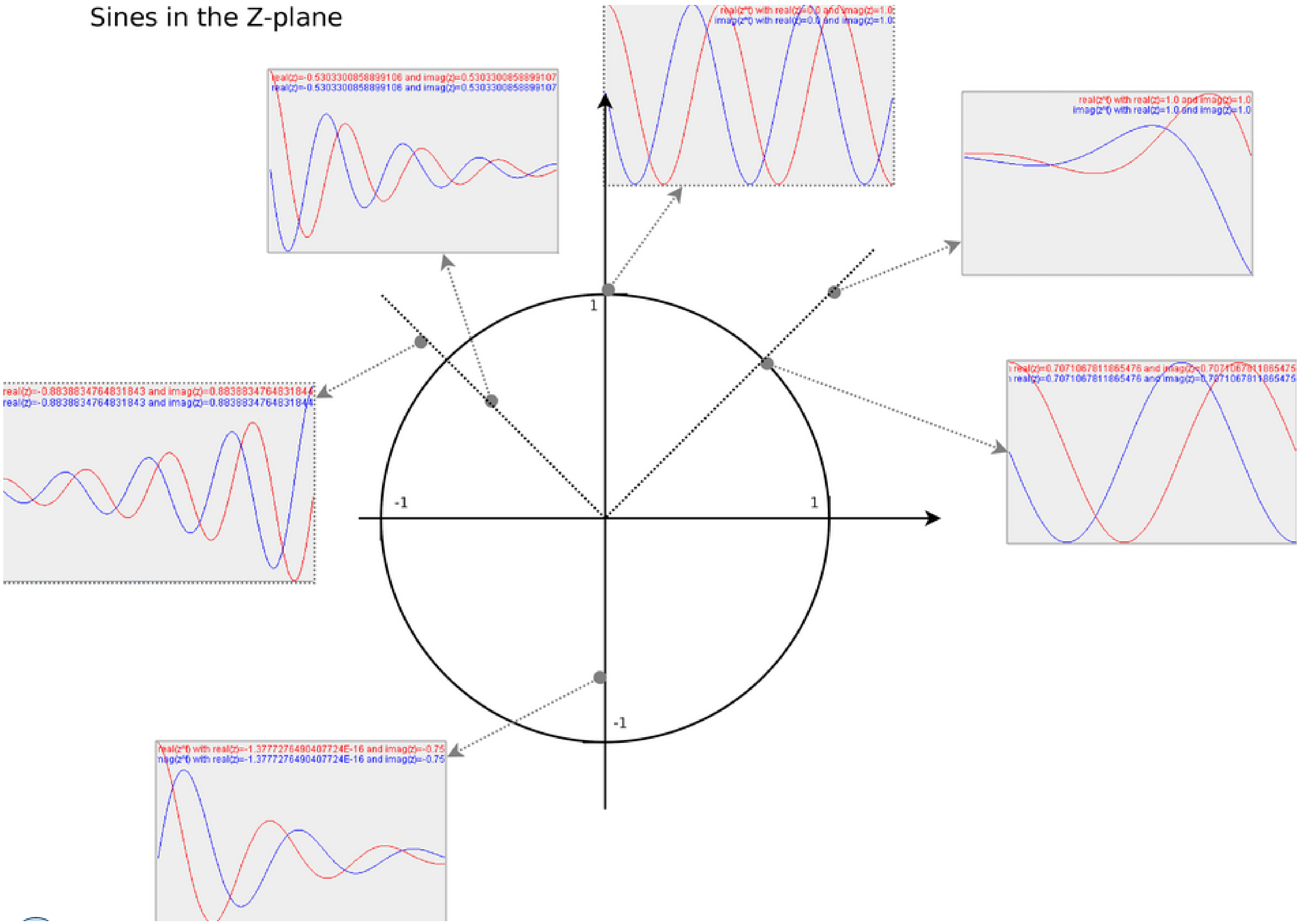
Polar Plane



Z-Transform vs. Laplace Transform



Z-Transform vs. Laplace Transform

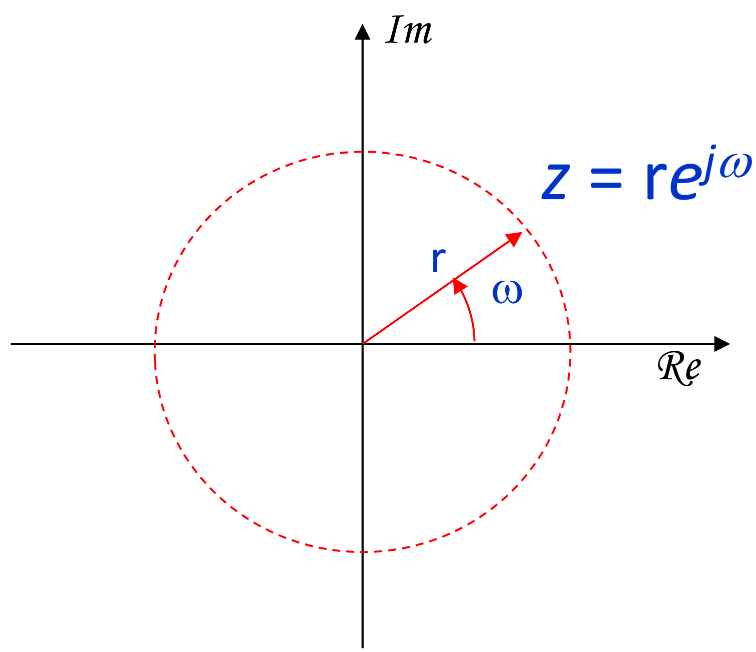


Direct Z-Transform: Definition

- The z-transform of sequence $x(n)$ is defined by

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Sum of an infinite series !



It will converge to some finite value only for certain range of values of z

Region of Convergence: ROC

Give a sequence $x(n)$, the set of values of z for which the z-transform converges, i.e., $|X(z)|<\infty$, is called the region of convergence.

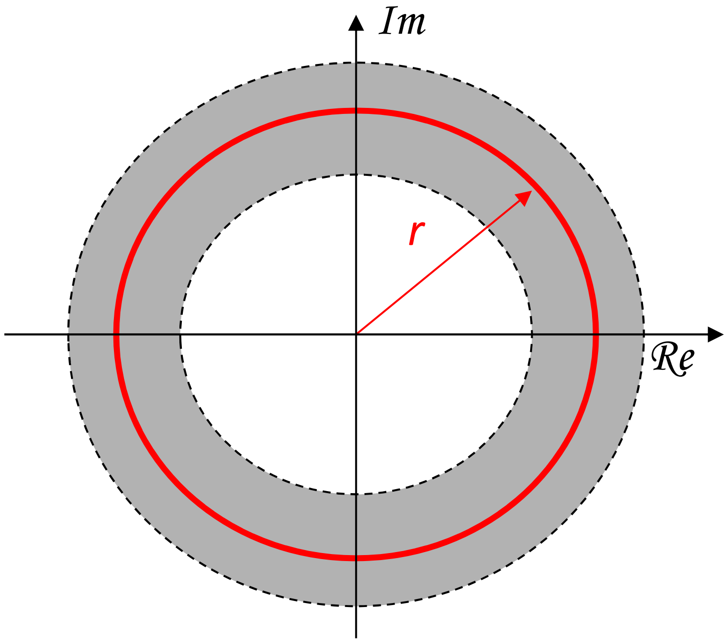
$$|X(z)| = \left| \sum_{n=-\infty}^{\infty} x(n) z^{-n} \right| = \sum_{n=-\infty}^{\infty} |x(n)||z|^{-n} < \infty$$

ROC is centered on origin and consists of a set of rings.

Region of Convergence: ROC

Example: ROC

$$|X(z)| = \left| \sum_{n=-\infty}^{\infty} x(n) z^{-n} \right| = \sum_{n=-\infty}^{\infty} |x(n)||z|^{-n} < \infty$$



ROC is an annual ring centered on the origin.

$$R_{x-} < |z| < R_{x+}$$

$$ROC = \{z = re^{j\omega} \mid R_{x-} < r < R_{x+}\}$$

Region of Convergence: ROC

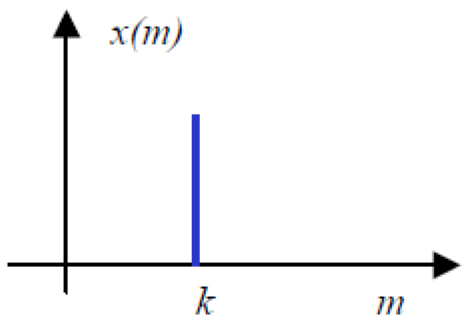
- Direct z-Transform: $$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$
- Notation: $$X(z) \equiv \mathcal{Z}\{x(n)\}$$

$$x(n) \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$$

- The z-Transform is, therefore, uniquely characterized by:
  1. expression for $X(z)$
  2. ROC of $X(z)$

Direct Z-Transform and ROC

Example: Right sided finite duration sequence



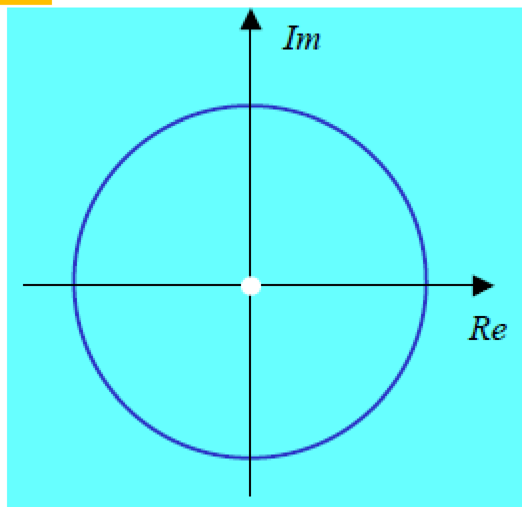
$$x(m) = \delta(m-k) = \begin{cases} 1 & m = k \\ 0 & m \neq k \end{cases}$$

$$\boxed{X(z) = \sum_{m=-\infty}^{\infty} \delta(m-k) z^{-m} = z^{-k}}$$

$$\boxed{ROC = ?}$$

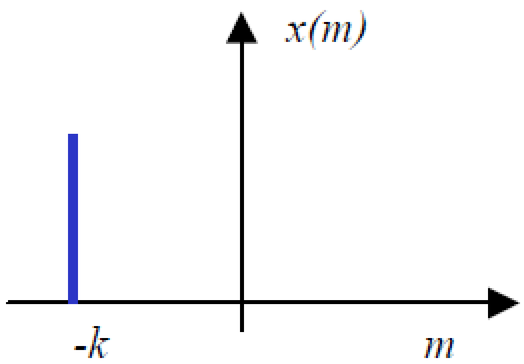
If you put *z=0*, what happens to *X(z)*?

$$\boxed{ROC = \text{Entire z-plane except } z = 0}$$



Direct Z-Transform and ROC

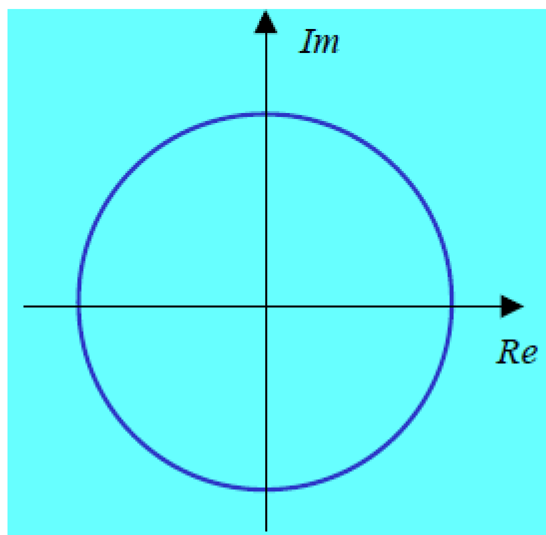
Example: Left sided finite duration sequence



$$x(m) = \delta(m+k) = \begin{cases} 1 & m = -k \\ 0 & m \neq -k \end{cases}$$

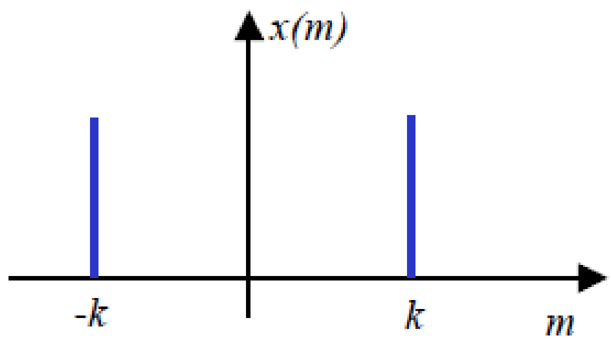
$$\boxed{X(z) = \sum_{m=-\infty}^{\infty} x(m) z^{-m} = z^{k}}$$

$$\boxed{ROC = \text{Entire z-plane except } z = \infty}$$



Direct Z-Transform and ROC

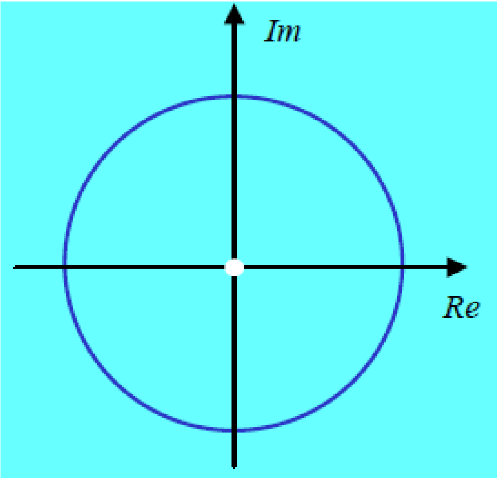
Example: Two sided finite duration sequence



$$x(m) = \delta(m+k) + \delta(m-k) = \begin{cases} 1 & m = \pm k \\ 0 & m \neq \pm k \end{cases}$$

$$\boxed{X(z) = \sum_{m=-\infty}^{\infty} x(m) z^{-m} = z^{k} + z^{-k}}$$

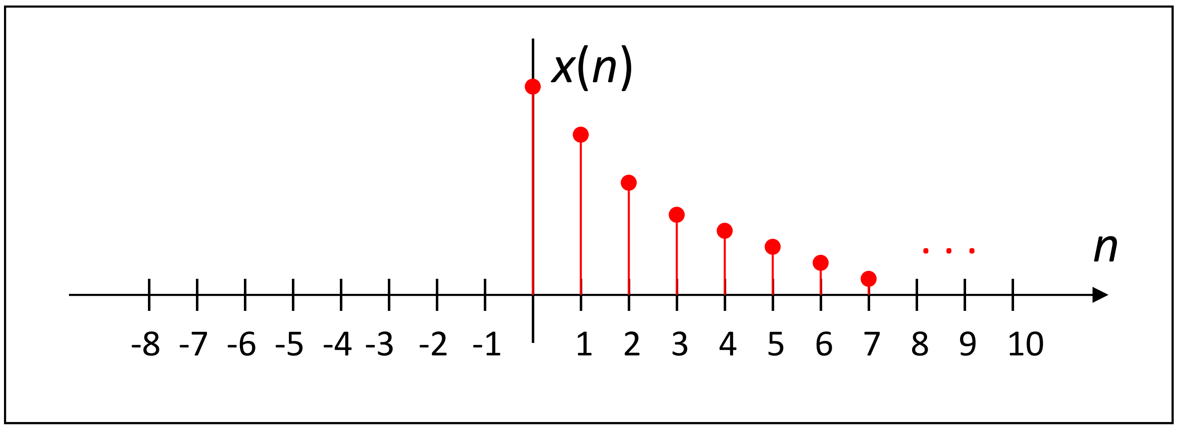
$$\boxed{ROC = \text{Entire z-plane except } z = \infty \text{ and } z = 0}$$



Direct Z-Transform and ROC

Example: A right sided infinite duration sequence

$$x(n) = a^n u(n)$$



Direct Z-Transform and ROC

Example: A right sided infinite duration sequence

$$\boxed{x(n) = a^n u(n)}$$

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

For convergence of $X(z)$, we require that

$$\sum_{n=0}^{\infty} |az^{-1}| < \infty \Rightarrow |az^{-1}| < 1$$

$$\Rightarrow |z| > |a|$$

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z-a}$$

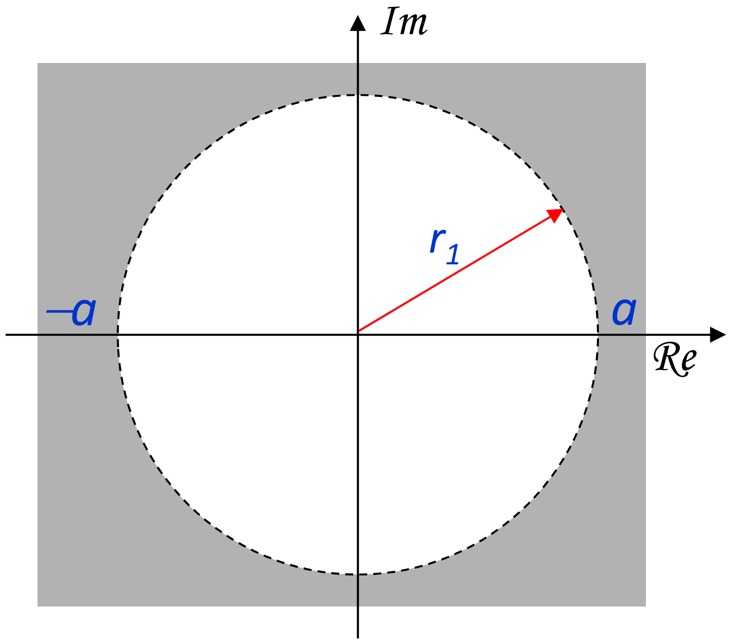
$$|z| > |a|$$

Direct Z-Transform and ROC

Example: A right sided infinite duration sequence

$$x(n) = a^n u(n)$$

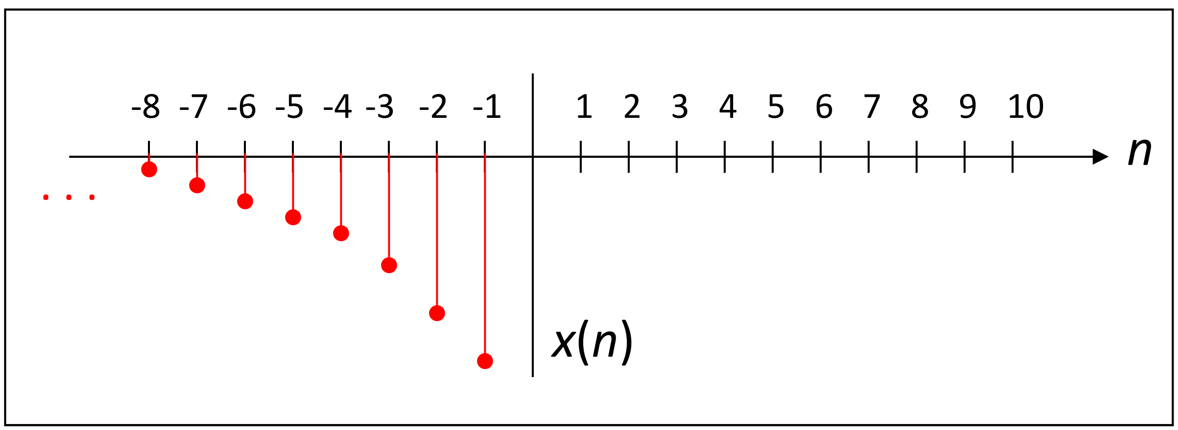
$$X(z) = \frac{z}{z-a}, \qquad |z| > |a|$$



Direct Z-Transform and ROC

Example: A left sided infinite duration sequence

$$x(n) = -a^n u(-n-1)$$



Direct Z-Transform and ROC

Example: A left sided infinite duration sequence

$$\boxed{x(n) = -a^n u(-n-1)}$$

$$X(z) = -\sum_{n=-\infty}^{\infty} a^n u(-n-1) z^{-n}$$

$$= -\sum_{n=-\infty}^{-1} a^n z^{-n}$$

$$= -\sum_{n=1}^{\infty} a^{-n} z^n$$

$$= 1 - \sum_{n=0}^{\infty} a^{-n} z^n$$

For convergence of $X(z)$, we require that

$$\sum_{n=0}^{\infty} |a^{-1} z| < \infty \Rightarrow |a^{-1} z| < 1$$

$$\Rightarrow |z| < |a|$$

$$X(z) = 1 - \sum_{n=0}^{\infty} (a^{-1} z)^n = 1 - \frac{1}{1 - a^{-1} z} = \frac{z}{z-a}$$

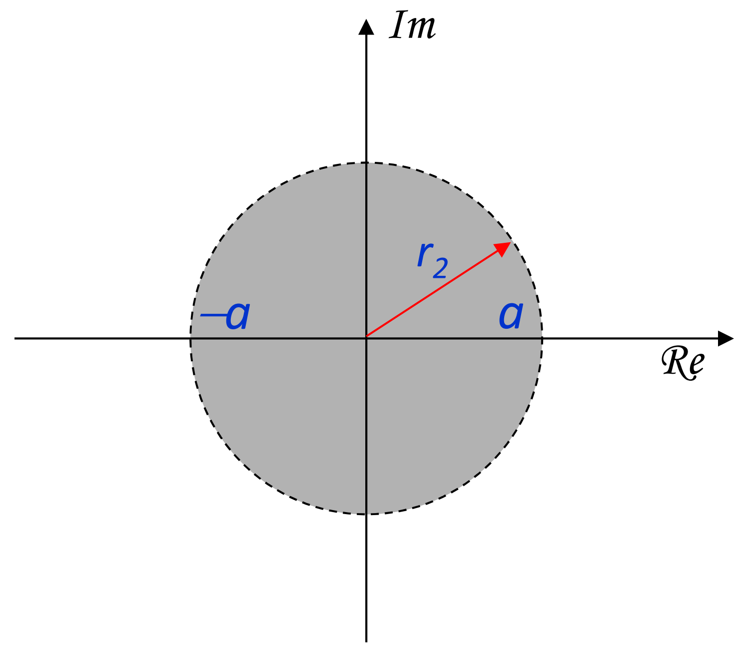
$$|z| < |a|$$

Direct Z-Transform and ROC

Example: A left sided infinite duration sequence

$$x(n) = -a^n u(-n-1)$$

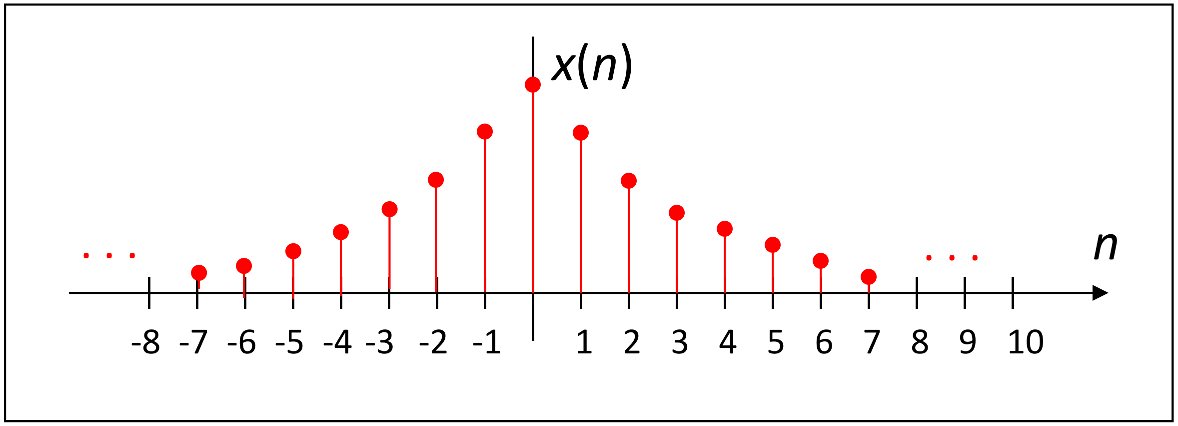
$$X(z) = \frac{z}{z-a}, \qquad |z| < |a|$$



Direct Z-Transform and ROC

Example: Two sided infinite duration sequence

$$x(n) = a^n u(n) + b^n u(-n-1)$$



Direct Z-Transform and ROC

Example: Two sided infinite duration sequence

$$x(n) = a^n u(n) + b^n u(-n-1)$$

$$X(z) = \sum_{n=-\infty}^{-1} b^n z^{-n} + \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=-\infty}^{-1} \left(bz^{-1}\right)^n + \sum_{n=0}^{\infty} \left(az^{-1}\right)^n$$

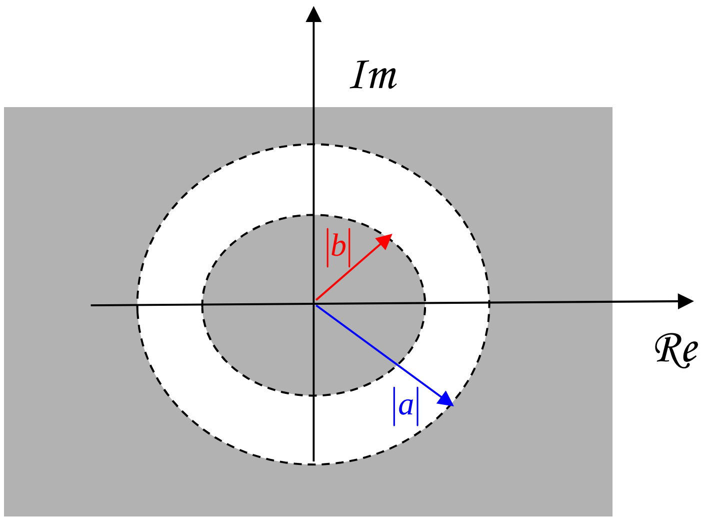
converges for $|z| < |b|$

converges for $|z| > |a|$

Two Cases:

$|b| < |a|$:

ROC's do not overlap. We cannot find values of z for which both series converge simultaneously.



Direct Z-Transform and ROC

Example: Two sided infinite duration sequence

$$\boxed{x(n) = a^n u(n) + b^n u\left(-n-1\right)}$$

$$X(z) = \sum_{n=-\infty}^{-1} b^n z^{-n} + \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=-\infty}^{-1} \left(bz^{-1}\right)^n + \sum_{n=0}^{\infty} \left(az^{-1}\right)^n$$

converges for $|z| < |b|$

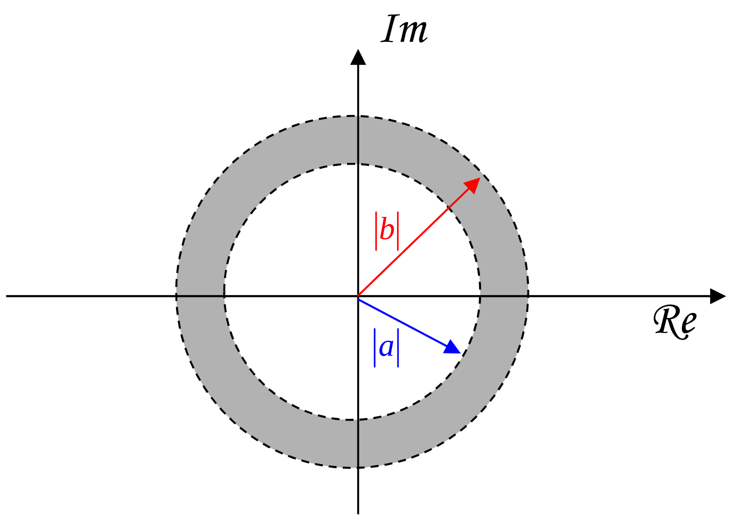
converges for $|z| > |a|$

Two Cases:

$|b| > |a|$:

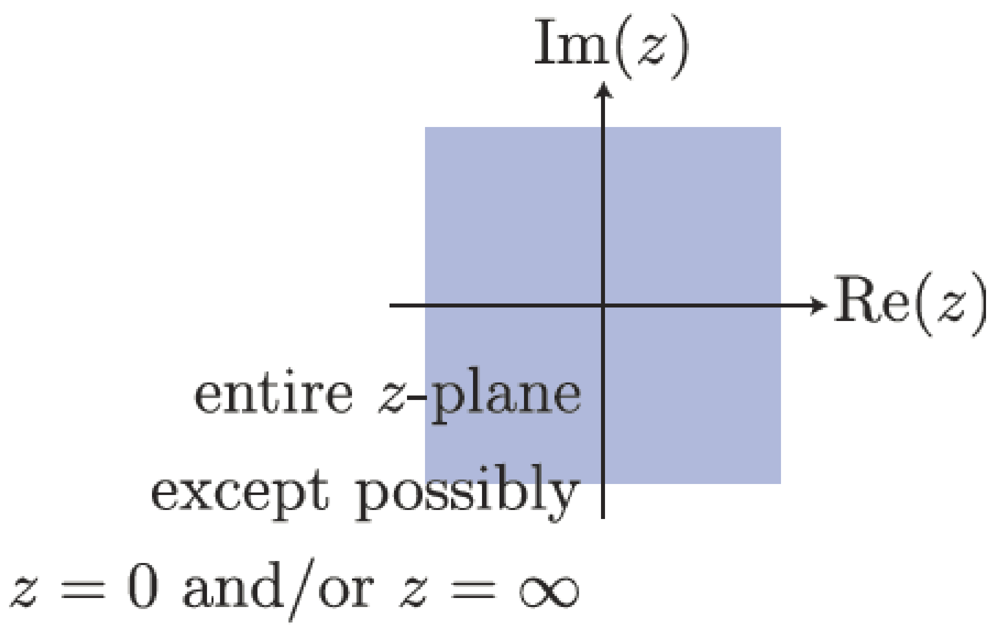
$$X(z) = \frac{1}{1-az^{-1}} - \frac{1}{1-bz^{-1}}$$

$$ROC: \quad |a| < |z| < |b|$$



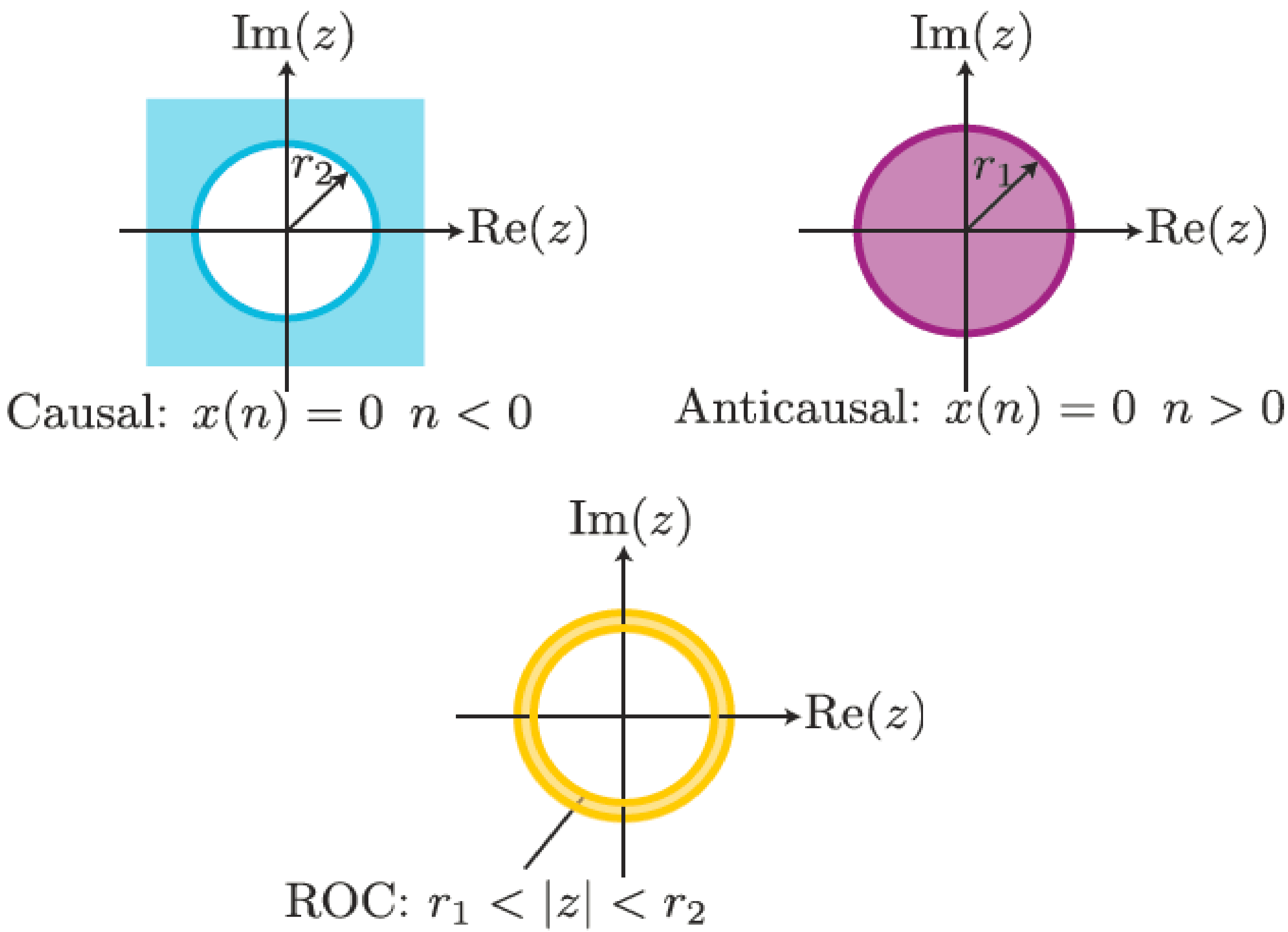
ROC Families

ROC Families: Finite Duration Signals



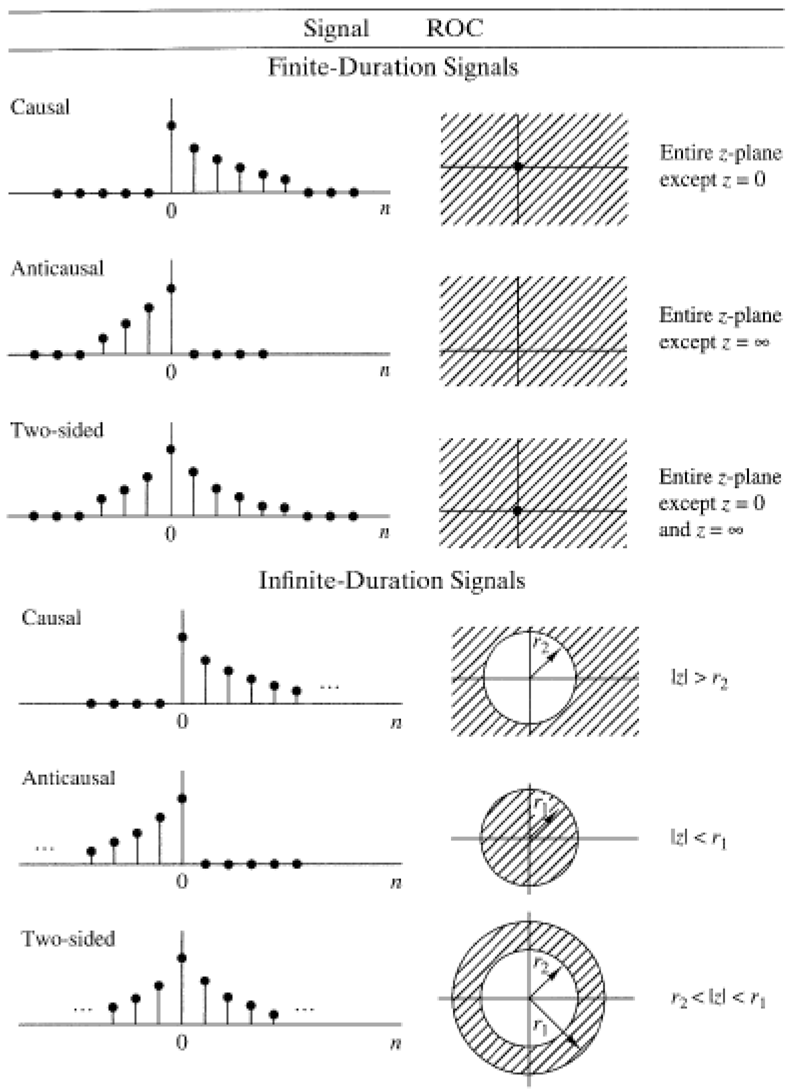
ROC Families

ROC Families: Infinite Duration Signals



Two-sided = Causal + Anticausal

ROC Families



Z-Transform Pairs

Sequence	z-Transform	ROC
$\delta(n)$	1	All z
$\delta(n-m)$	z^{-m}	All z except 0 (if m>0) or ∞ (if m<0)
$u(n)$	$\dfrac{1}{1-z^{-1}}$	$\lvert z\rvert>1$
$-u(-n-1)$	$\dfrac{1}{1-z^{-1}}$	$\lvert z\rvert<1$
$a^n u(n)$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert>\lvert a\rvert$
$-a^n u(-n-1)$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert<\lvert a\rvert$

Z-Transform Pairs

Sequence	z-Transform	ROC
$[\cos\omega_0 n]u(n)$	$\dfrac{1-[\cos\omega_0]z^{-1}}{1-[2\cos\omega_0]z^{-1}+z^{-2}}$	$\|z\|>1$
$[\sin\omega_0 n]u(n)$	$\dfrac{[\sin\omega_0]z^{-1}}{1-[2\cos\omega_0]z^{-1}+z^{-2}}$	$\|z\|>1$
$[r^n\cos\omega_0 n]u(n)$	$\dfrac{1-[r\cos\omega_0]z^{-1}}{1-[2r\cos\omega_0]z^{-1}+r^2z^{-2}}$	$\|z\|>r$
$[r^n\sin\omega_0 n]u(n)$	$\dfrac{[r\sin\omega_0]z^{-1}}{1-[2r\cos\omega_0]z^{-1}+r^2z^{-2}}$	$\|z\|>r$
$\begin{cases} a^n & 0\le n\le N-1 \\ 0 & \text{otherwise}\end{cases}$	$\dfrac{1-a^Nz^{-N}}{1-az^{-1}}$	$\|z\|>0$

Z-Transform Pairs

YOU WILL NOT BE PROVIDED ANY KIND OF Z-TRANSFORM PAIR TABLE/FORMULAE IN ANY QUIZ OR ANY EXAM

Week 4: Z-Transform

- Direct Z-Transform and ROC
- **Z-Transform Properties**
- Rational Z-Transform
- Inversion of Z-Transform
- Analysis of LTI systems in Z-domain

Z-Transform Properties

Linearity:

$$
\begin{aligned}
x_1(n) &\overset{z}{\longleftrightarrow} X_1(z), \mathrm{ROC}_1 \\
x_2(n) &\overset{z}{\longleftrightarrow} X_2(z), \mathrm{ROC}_2
\end{aligned}
$$

$$
\begin{aligned}
a_1x_1(n) + a_2x_2(n) \overset{z}{\longleftrightarrow}\ & a_1X_1(z) + a_2X_2(z), \\
& \text{At least } \mathrm{ROC}_1 \cap \mathrm{ROC}_2
\end{aligned}
$$

Z-Transform Properties

Linearity: Proof

Let $x(n) = a_1 x_1(n) + a_2 x_2(n)$

$$
\begin{aligned}
X(z) &= \sum_{n=-\infty}^{\infty} x(n) z^{-n} \\
&= \sum_{n=-\infty}^{\infty} (a_1 x_1(n) + a_2 x_2(n)) z^{-n} \\
&= a_1 \sum_{n=-\infty}^{\infty} x_1(n) z^{-n} + a_2 \sum_{n=-\infty}^{\infty} x_2(n) z^{-n} \\
&= a_1 X_1(z) + a_2 X_2(z)
\end{aligned}
$$

ROC: At least $\mathrm{ROC}_1 \cap \mathrm{ROC}_2$

Z-Transform Properties

Linearity: Example:

$$x(n) = [3(2^n) - 4(3^n)]u(n)$$

$$x_1(n) = 2^n u(n)$$

$$x(n) = 3x_1(n) - 4x_2(n)$$

$$x_2(n) = 3^n u(n)$$

$$X(z) = 3X_1(z) - 4X_2(z)$$

$$x_1(n) = 2^n u(n) \stackrel{z}{\longleftrightarrow} X_1(z) = \frac{1}{1 - 2z^{-1}}, \qquad \text{ROC: } |z| > 2$$

$$x_2(n) = 3^n u(n) \stackrel{z}{\longleftrightarrow} X_2(z) = \frac{1}{1 - 3z^{-1}}, \qquad \text{ROC: } |z| > 3$$

$$\boxed{X(z) = \frac{3}{1 - 2z^{-1}} - \frac{4}{1 - 3z^{-1}}, \qquad \text{ROC: } |z| > 3}$$

Z-Transform Properties

Time Shift:

$$x(n) \stackrel{z}{\longleftrightarrow} X(z), \qquad \text{ROC}$$

$$x(n-k) \stackrel{z}{\longleftrightarrow} z^{-k}X(z), \qquad \text{At least ROC}$$

except $z = 0$ $(k > 0)$ or

$z = \infty$ $(k < 0)$

Z-Transform Properties

Time Shift: Proof

Let $y(n) = x(n-k)$.

$$
\begin{aligned}
Y(z) &= \sum_{n=-\infty}^{\infty} y(n) z^{-n} \\
&= \sum_{n=-\infty}^{\infty} (x(n-k)) z^{-n} \\
&\quad \text{Let } m = n - k \\
&= \sum_{m=-\infty}^{\infty} x(m) z^{-(m+k)} = \sum_{m=-\infty}^{\infty} x(m) z^{-m} z^{-k} \\
&= z^{-k} \left[\sum_{m=-\infty}^{\infty} x(m) z^{-m} \right] = z^{-k} X(z)
\end{aligned}
$$

Z-Transform Properties

Time Shift: ROC and Examples:

$$x(n) = \delta(n) \quad \stackrel{\mathcal{Z}}{\longleftrightarrow} \quad X(z) = 1, \text{ROC: } \underline{\text{entire } z\text{-plane}}$$

- Example: For $k = -1$

$$y(n) = x(n - (-1)) = x(n+1) = \delta(n+1)$$

$$y(n) = \delta(n+1) \quad \stackrel{\mathcal{Z}}{\longleftrightarrow} \quad Y(z) = z, \text{ROC: } \underline{\text{entire } z\text{-plane}}$$

$$\text{except } z = \infty$$

- Example: For $k = 1$

$$y(n) = x(n - (1)) = x(n-1) = \delta(n-1)$$

$$y(n) = \delta(n-1) \quad \stackrel{\mathcal{Z}}{\longleftrightarrow} \quad Y(z) = z^{-1}, \text{ROC: } \underline{\text{entire } z\text{-plane}}$$

$$\text{except } z = 0$$

Z-Transform Properties

Scaling in z-domain:

$$x(n) \overset{z}{\longleftrightarrow} X(z), \qquad \text{ROC: } r_1 < |z| < r_2$$

$$a^n x(n) \overset{z}{\longleftrightarrow} X(a^{-1}z), \qquad \text{ROC: } |a|r_1 < |z| < |a|r_2$$

Z-Transform Properties

Scaling in z-domain: Proof

Let $y(n) = a^n x(n)$

$$
\begin{aligned}
Y(z) &= \sum_{n=-\infty}^{\infty} y(n) z^{-n} \\
&= \sum_{n=-\infty}^{\infty} a^n x(n) z^{-n} \\
&= \sum_{n=-\infty}^{\infty} x(n) (a^{-1}z)^{-n} \\
&= X(a^{-1}z)
\end{aligned}
$$

ROC: $r_1 < |a^{-1}z| = \frac{|z|}{|a|} < r_2 \;\equiv\; |a|r_1 < |z| < |a|r_2$

Z-Transform Properties

Property	Time Domain	z-Domain	ROC
Notation:	$x(n)$	$X(z)$	ROC: $r_2 < \|z\| < r_1$
	$x_1(n)$	$X_1(z)$	ROC_1
	$x_2(n)$	$X_1(z)$	ROC_2
Linearity:	$a_1x_1(n) + a_2x_2(n)$	$a_1X_1(z) + a_2X_2(z)$	At least $ROC_1 \cap ROC_2$
Time shifting:	$x(n-k)$	$z^{-k}X(z)$	ROC, except $z = 0$ (if $k > 0$) and $z = \infty$ (if $k < 0$)
z-Scaling:	$a^n x(n)$	$X(a^{-1}z)$	$\|a\|r_2 < \|z\| < \|a\|r_1$
Time reversal	$x(-n)$	$X(z^{-1})$	$\frac{1}{r_1} < \|z\| < \frac{1}{r_2}$
Conjugation:	$x^*(n)$	$X^*(z^*)$	ROC
z-Differentiation:	$n\ x(n)$	$-z\frac{dX(z)}{dz}$	$r_2 < \|z\| < r_1$
Convolution:	$x_1(n) * x_2(n)$	$X_1(z)X_2(z)$	At least $ROC_1 \cap ROC_2$

among others ...

Z-Transform Properties

Convolution Property:

$$x(n) = x_1(n) * x_2(n) \quad \Longleftrightarrow \quad X(z) = X_1(z) \cdot X_2(z)$$

$$x_1(n) * x_2(n) = \sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k)$$

$$\mathcal{Z}[x_1(n) * x_2(n)] = \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x_1(k) x_2(n-k) \right) z^{-n}$$

$$= \sum_{k=-\infty}^{\infty} x_1(k) \sum_{n=-\infty}^{\infty} x_2(n-k) z^{-n} \qquad = \sum_{k=-\infty}^{\infty} x_1(k) z^{-k} \sum_{n=-\infty}^{\infty} x_2(n) z^{-n}$$

$$= X_1(z) X_2(z) \qquad ROC: \quad ROC_{x_1} \cap ROC_{x_2}$$

Week 4: Z-Transform

- Direct Z-Transform and ROC
- Z-Transform Properties
- **Rational Z-Transform**
- Inversion of Z-Transform
- Analysis of LTI systems in Z-domain

Rational Z-Transform

Why Rational?

- $X(z)$ is a rational function iff it can be represented as the ratio of two polynomials in z^{-1} (or z):

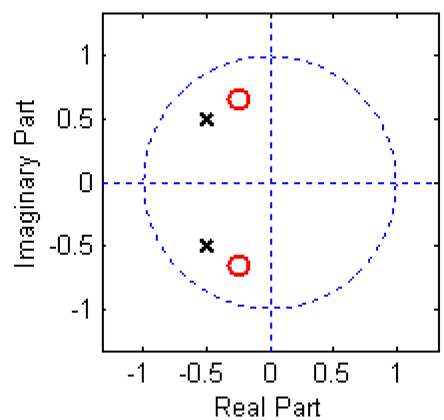
$$X(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}}$$

- For LTI systems that are represented by LCCDEs, the z-Transform of the unit sample response $h(n)$, denoted $H(z) = \mathcal{Z}\{h(n)\}$, is rational

Rational Z-Transform

Poles and Zeros

- zeros of $X(z)$: values of z for which $X(z) = 0$
- poles of $X(z)$: values of z for which $X(z) = \infty$



Rational Z-Transform

Poles and Zeros of the Rational z-Transform

Let $a_0, b_0 \neq 0$:

$$
\begin{aligned}
X(z) = \frac{B(z)}{A(z)} &= \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_N z^{-N}} \\
&= \left(\frac{b_0 z^{-M}}{a_0 z^{-N}}\right) \frac{z^M + (b_1/b_0) z^{M-1} + \cdots + b_M/b_0}{z^N + (a_1/a_0) z^{N-1} + \cdots + a_N/a_0} \\
&= \frac{b_0}{a_0} z^{-M+N} \frac{(z - z_1)(z - z_2) \cdots (z - z_M)}{(z - p_1)(z - p_2) \cdots (z - p_N)} \\
&= G z^{N-M} \frac{\prod_{k=1}^{M} (z - z_k)}{\prod_{k=1}^{N} (z - p_k)}
\end{aligned}
$$

Rational Z-Transform

$$X(z) = G z^{N-M} \frac{\prod_{k=1}^{M}(z - z_k)}{\prod_{k=1}^{N}(z - p_k)}$$

- $X(z)$ has M finite zeros at $z = z_1, z_2, \ldots, z_M$
- $X(z)$ has N finite poles at $z = p_1, p_2, \ldots, p_N$

- For $N - M \neq 0$
  - if $N - M > 0$, there are $|N - M|$ zeros at origin, $z = 0$
  - if $N - M < 0$, there are $|N - M|$ poles at origin, $z = 0$

There can also be poles or zeros at $z = \infty$, depending if $X(\infty) = \infty$ or $X(\infty) = 0$
Counting all of the above, there will be the same number of poles and zeros.

$$\boxed{\text{Total number of zeros} = \text{Total number of poles}}$$

Rational **Z-Transform**

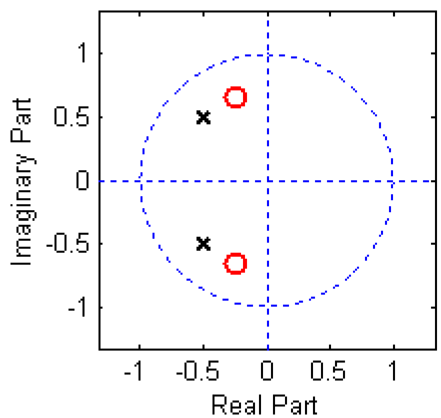
$$X(z) = G z^{N-M} \frac{\prod_{k=1}^{M}(z - z_k)}{\prod_{k=1}^{N}(z - p_k)}$$

$X(z)$ is *completely determined by is pole-zero locations* up to the scale factor G. The scale factor only affects the *amplitude* (or units) of the signal or system, whereas the poles and zeros affect the *behavior*.

A **pole-zero plot** is a graphic description of rational $X(z)$

Use $\circ$ for zeros and $\times$ for poles.

Multiple poles or zeros indicated with adjacent number.



Rational **Z-Transform**

Example:

Example:

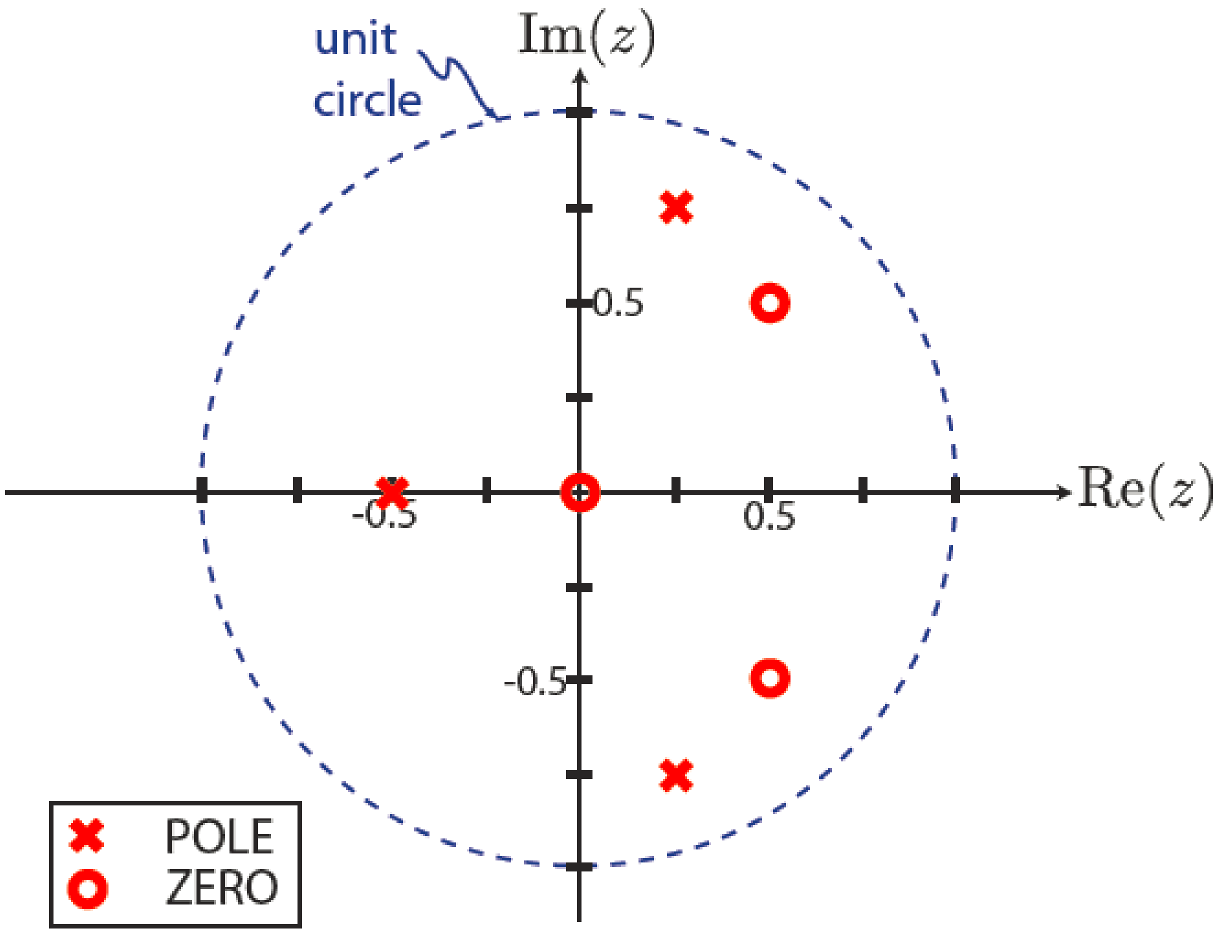
$$\begin{aligned} X(z) &= z\frac{2z^2-2z+1}{16z^3+6z+5} \\ &= (z-0)\frac{(z-(\frac{1}{2}+j\frac{1}{2}))(z-(\frac{1}{2}-j\frac{1}{2}))}{(z-(\frac{1}{4}+j\frac{3}{4}))(z-(\frac{1}{4}-j\frac{3}{4}))(z-(-\frac{1}{2}))} \end{aligned}$$

$$\boxed{\text{poles: } z=\frac{1}{4}\pm j\frac{3}{4}, -\frac{1}{2}}$$

$$\boxed{\text{zeros: } z=0, \frac{1}{2}\pm j\frac{1}{2}}$$

Rational Z-Transform

Example: $\boxed{\text{poles: } z = \frac{1}{4} \pm j\frac{3}{4}, -\frac{1}{2}}$, $\boxed{\text{zeros: } z = 0, \frac{1}{2} \pm j\frac{1}{2}}$

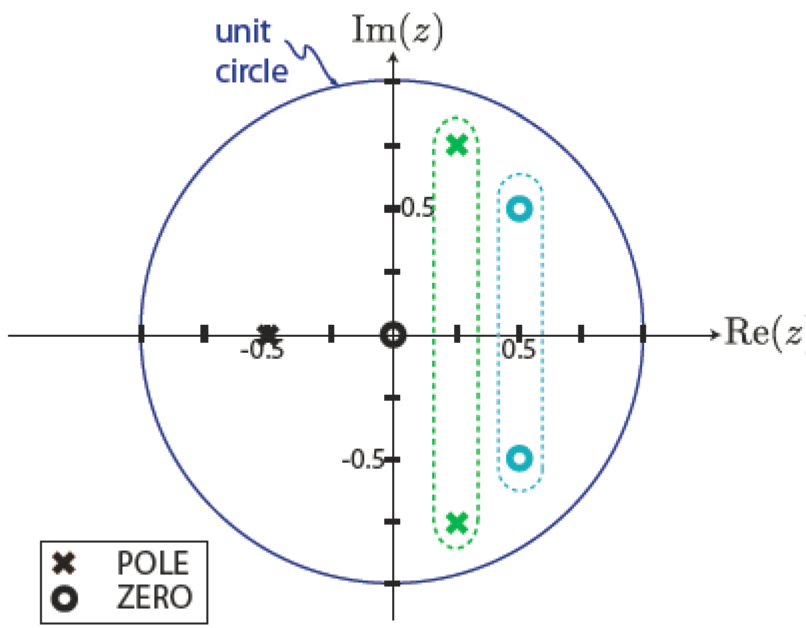


Rational Z-Transform

Pole-Zero Plot and Conjugate Pairs

- For real time-domain signals, the coefficients of $X(z)$ are necessarily real
  - complex poles and zeros must occur in conjugate pairs
  - note: real poles and zeros do not have to be paired up

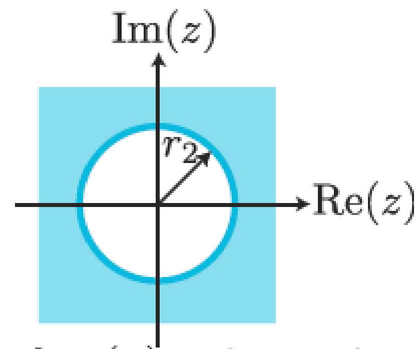
$$X(z) = z\frac{2z^2 - 2z + 1}{16z^3 + 6z + 5} \Longrightarrow$$



Rational **Z-Transform**

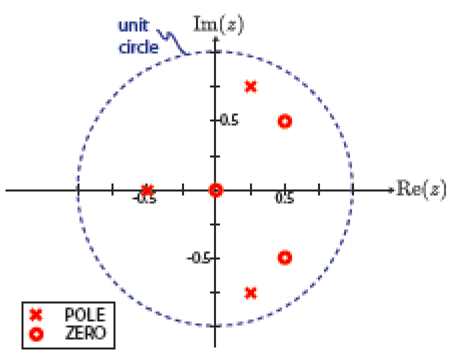
Pole-Zero Plot and the ROC

- Recall, for causal signals, the ROC will be the outer region of a disk



Causal: $x(n) = 0 \;\; n < 0$

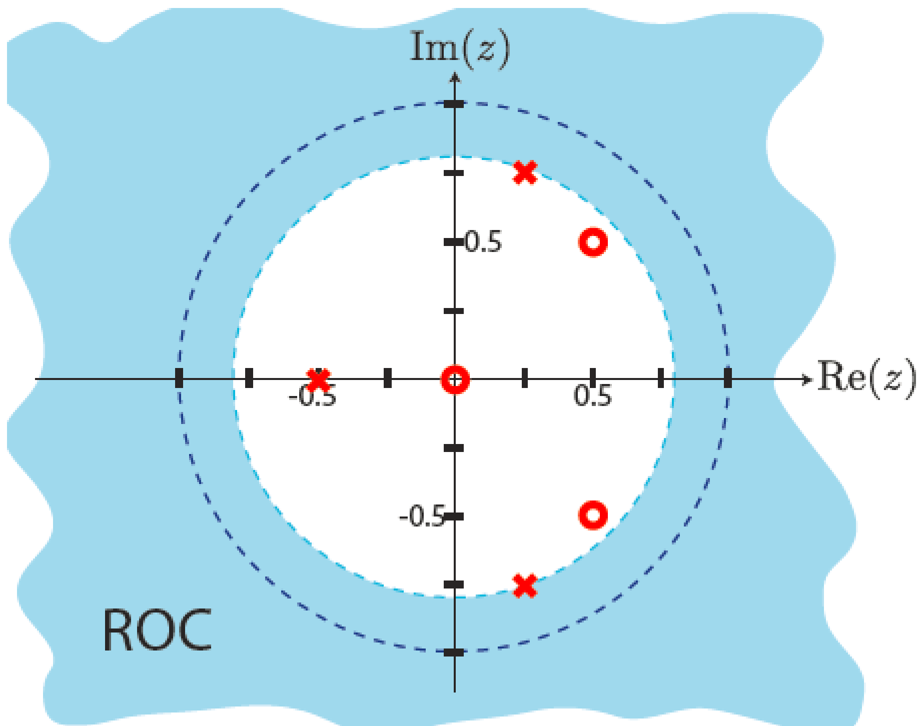
- ROC cannot necessarily include poles $(\because X(p_k) = \infty)$



Rational **Z-Transform**

Pole-Zero Plot and the ROC

- Therefore, for a causal signal the ROC is the smallest (origin-centered) circle encompassing all the poles.



Rational Z-Transform

Causality and Stability

- Recall,

$$\text{LTI system is stable} \iff \sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

- Moreover,

$$\begin{aligned} |H(z)| &= \left| \sum_{n=-\infty}^{\infty} h(n) z^{-n} \right| \le \sum_{n=-\infty}^{\infty} |h(n) z^{-n}| \\ &= \sum_{n=-\infty}^{\infty} |h(n)| \qquad \text{for } |z| = 1 \end{aligned}$$

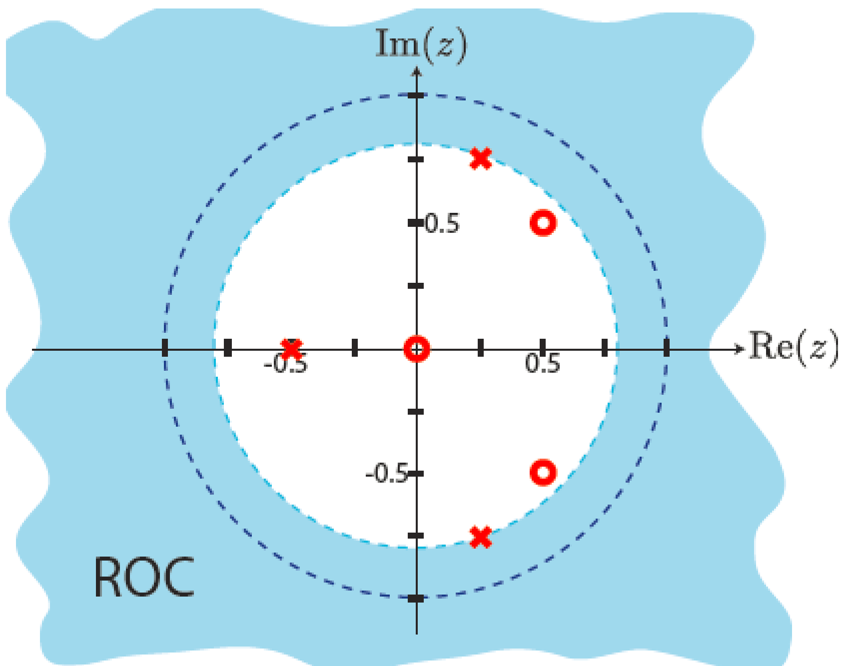
- It can be shown:

$$\text{LTI system is stable} \iff \sum_{n=-\infty}^{\infty} |h(n)| < \infty \iff \text{ROC of } H(z) \text{ contains unit circle}$$

Rational Z-Transform

Pole-Zero Plot, Causality and Stability

- For stable systems, the ROC will include the unit circle.

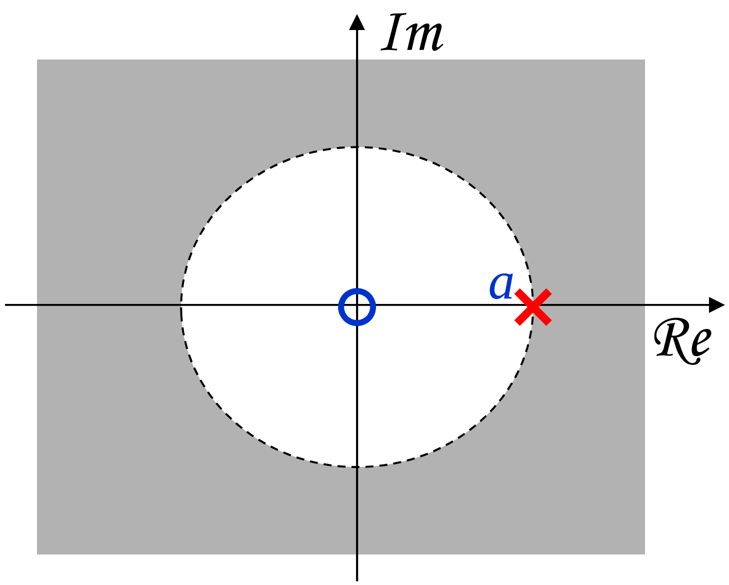


- For stability of a causal system, the poles must lie inside the unit circle.

Rational Z-Transform

Example: A right sided Sequence

$$x(n) = a^n u(n) \quad \Rightarrow \quad X(z) = \frac{z}{z-a}, \qquad |z| > |a|$$

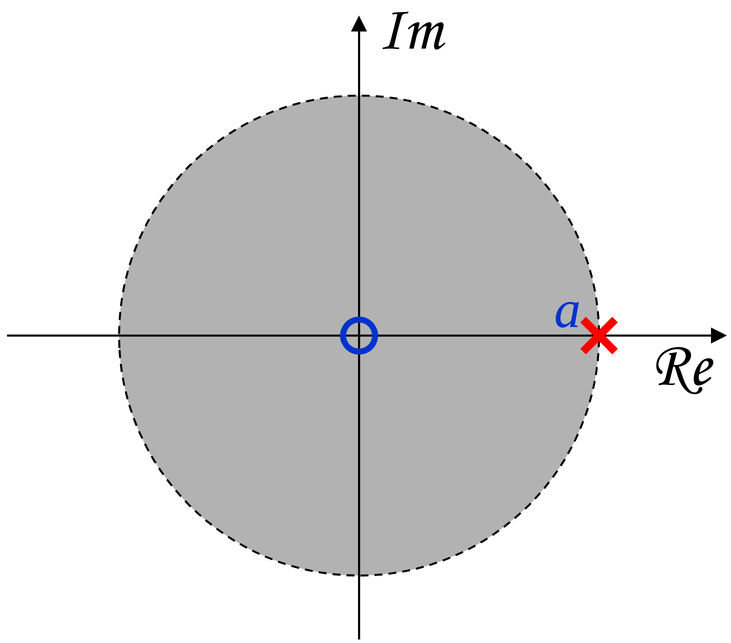


ROC is bounded by the pole and is the exterior of a circle.

Rational **Z-Transform**

Example: A left sided Sequence

$$x(n) = -a^n u(-n-1) \Rightarrow X(z) = \frac{z}{z-a}, \qquad |z| < |a|$$



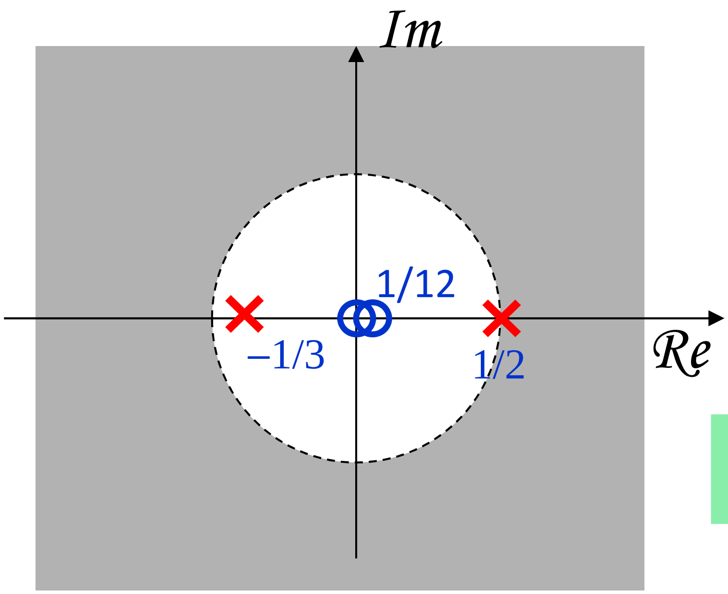
ROC is bounded by the pole and is the interior of a circle.

Rational **Z-Transform**

Example: Sum of Two Right Sided Sequences

$$x(n) = (\tfrac{1}{2})^n u(n) + (-\tfrac{1}{3})^n u(n)$$

$$\Rightarrow \quad X(z) = \frac{z}{z-\frac{1}{2}} + \frac{z}{z+\frac{1}{3}} = \frac{2z(z-\frac{1}{12})}{(z-\frac{1}{2})(z+\frac{1}{3})}$$



ROC is bounded by poles and is the exterior of a circle.

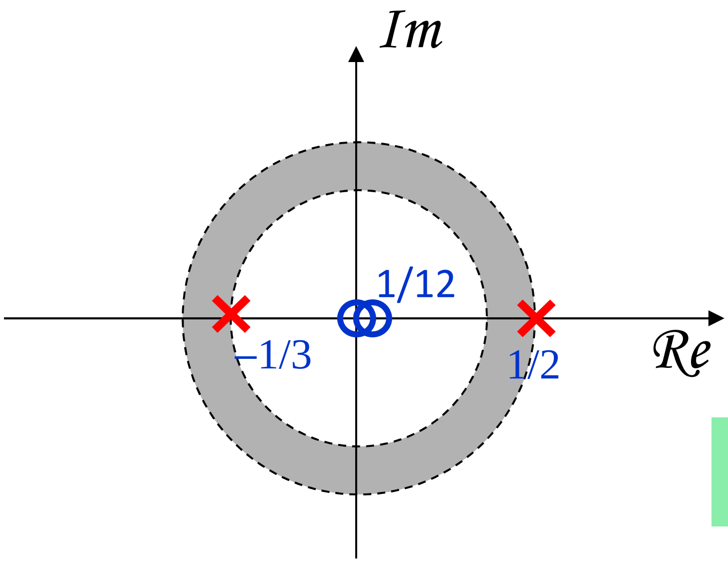
ROC does not include any pole.

Rational **Z-Transform**

Example: A Two Sided Sequence

$$x(n) = (-\tfrac{1}{3})^n u(n) - (\tfrac{1}{2})^n u(-n-1)$$

$$\Rightarrow \quad X(z) = \frac{z}{z+\frac{1}{3}} + \frac{z}{z-\frac{1}{2}} = \frac{2z(z-\frac{1}{12})}{(z+\frac{1}{3})(z-\frac{1}{2})}$$



ROC is bounded by poles and is a ring.

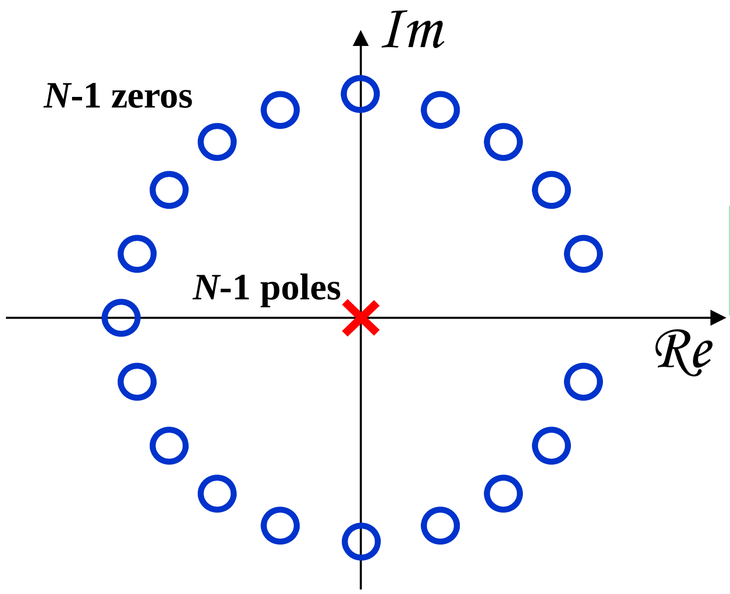
ROC does not include any pole.

Rational **Z-Transform**

Example: A Finite Sequence

$$x(n) = a^n, \qquad 0 \le n \le N-1$$

$$X(z) = \sum_{n=0}^{N-1} a^n z^{-n} = \sum_{n=0}^{N-1} (az^{-1})^n = \frac{1-(az^{-1})^N}{1-az^{-1}} = \frac{1}{z^{N-1}} \frac{z^N - a^N}{z-a}$$



ROC: $0 < z < \infty$

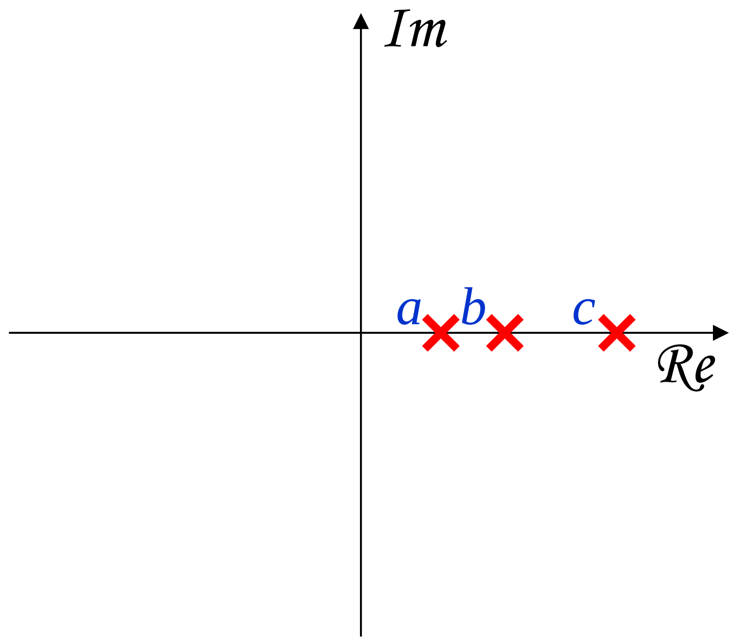
ROC does not include any pole.

Always Stable

Rational **Z-Transform**

Example: Find the possible ROC's

Consider the rational z-transform with the pole pattern:

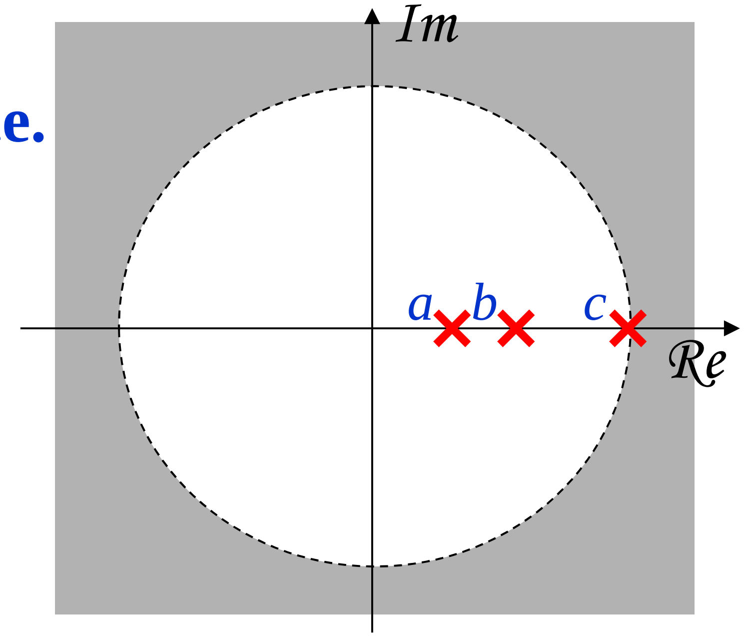


Rational **Z-Transform**

Example: Find the possible ROC's

Consider the rational z-transform with the pole pattern:

Case 1: A right sided Sequence.

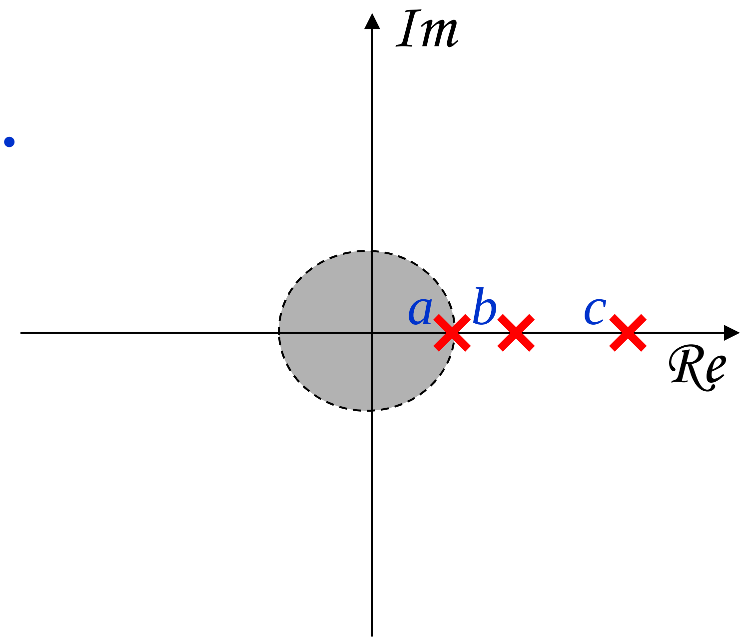


Rational **Z-Transform**

Example: Find the possible ROC's

Consider the rational z-transform with the pole pattern:

Case 2: A left sided Sequence.

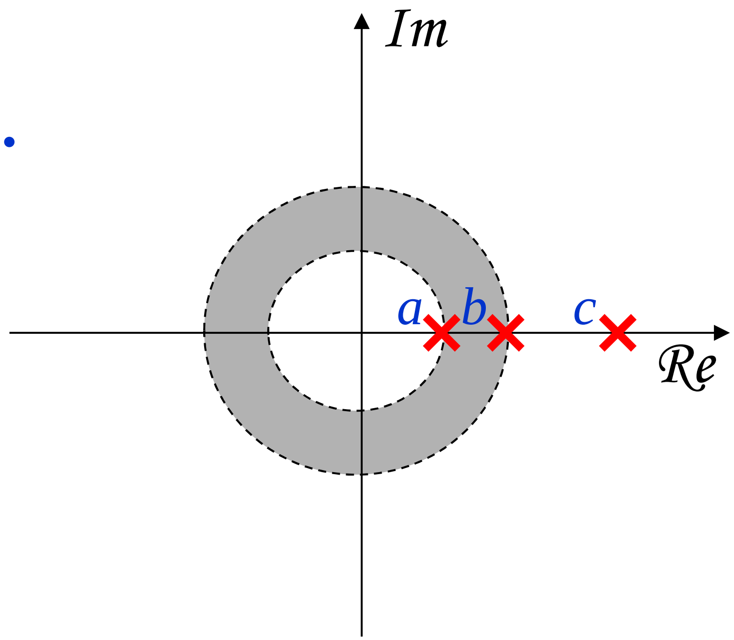


Rational **Z-Transform**

Example: Find the possible ROC's

Consider the rational z-transform with the pole pattern:

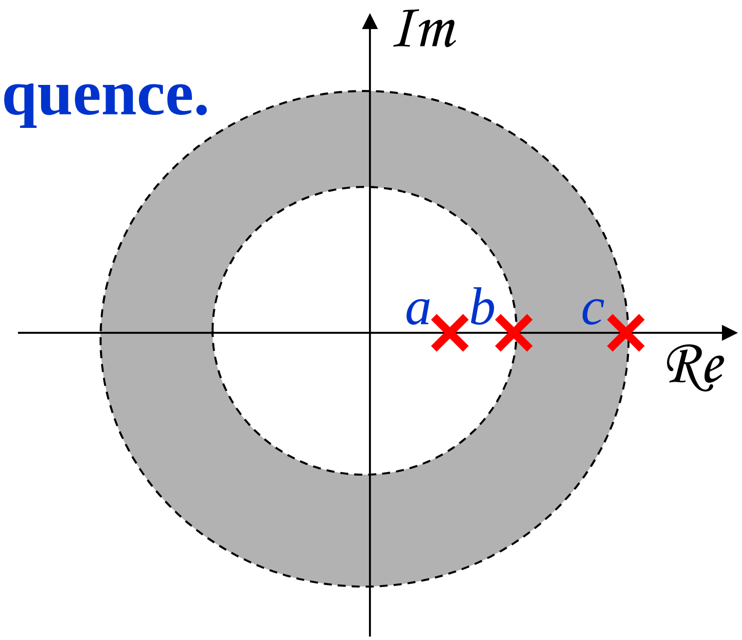
Case 3: A two sided Sequence.



Rational **Z-Transform**

Example: Find the possible ROC's

Case 4: Another two sided Sequence.

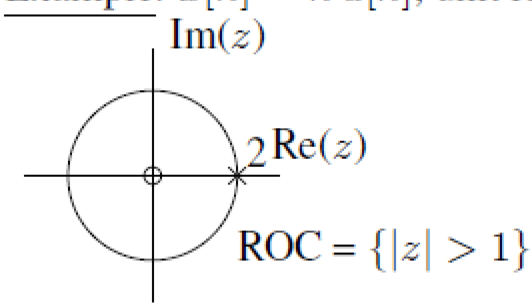


Rational Z-Transform

Rational Z –Transform: Outcome

- **Go from *x(n)* to *X(z)* to pole zero plot**

Example. $x[n] = n\,u[n]$, unit ramp signal. Previously showed that $X(z) = \frac{z^{-1}}{(1-z^{-1})^2} = \frac{z}{(z-1)^2},\ \{|z| > 1\}$.

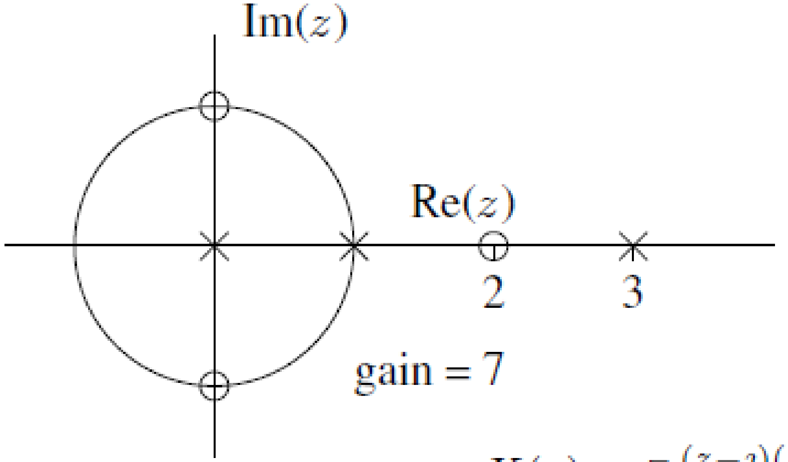


Rational **Z-Transform**

Rational Z –Transform: Outcome

- **Go from pole zero plot to *X(z)* to *x(n)***

Example. What are possible ROC's in following case? Answer: $\{|z| < 1\}$, $\{1 < |z| < 3\}$, or $\{3 < |z|\}$



$$X(z) = 7\frac{(z-j)(z+j)(z-2)}{(z-0)(z-1)(z-3)} = 7\frac{(1-jz^{-1})(1+jz^{-1})(1-2z^{-1})}{(1-z^{-1})(1-3z^{-1})}.$$ But what is $x[n]$?

Inverse Z-Transform: coming soon !

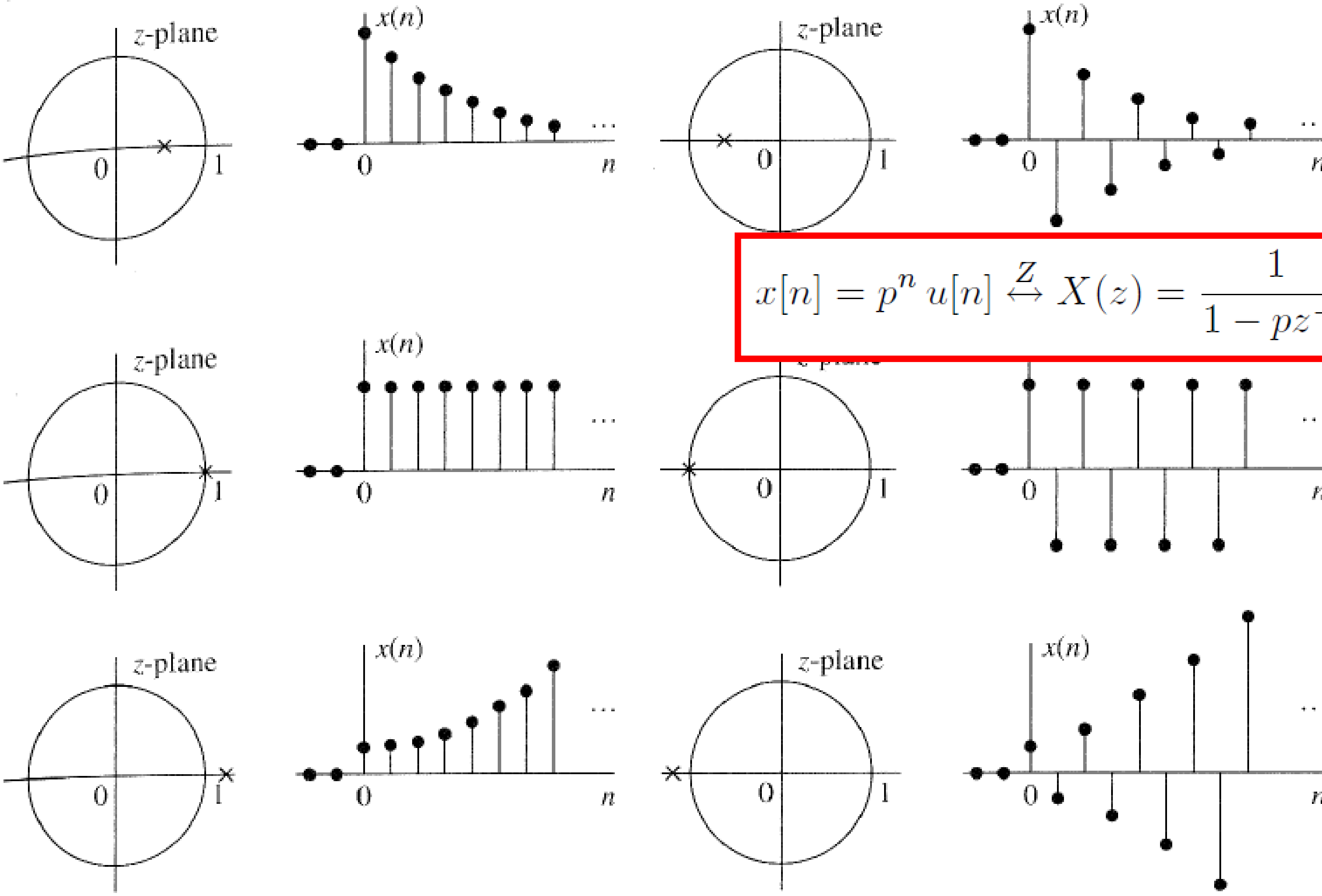
Rational **Z-Transform**

Pole Location and Time domain Behavior of real Causal Signals

Single Real pole:

$$\boxed{x[n] = p^n\, u[n] \overset{Z}{\longleftrightarrow} X(z) = \frac{1}{1 - pz^{-1}} = \frac{z}{z - p}.}$$

Rational **Z-Transform**



$$x[n] = p^n\, u[n] \overset{Z}{\leftrightarrow} X(z) = \frac{1}{1 - pz^{-1}} = \frac{z}{z - p}.$$

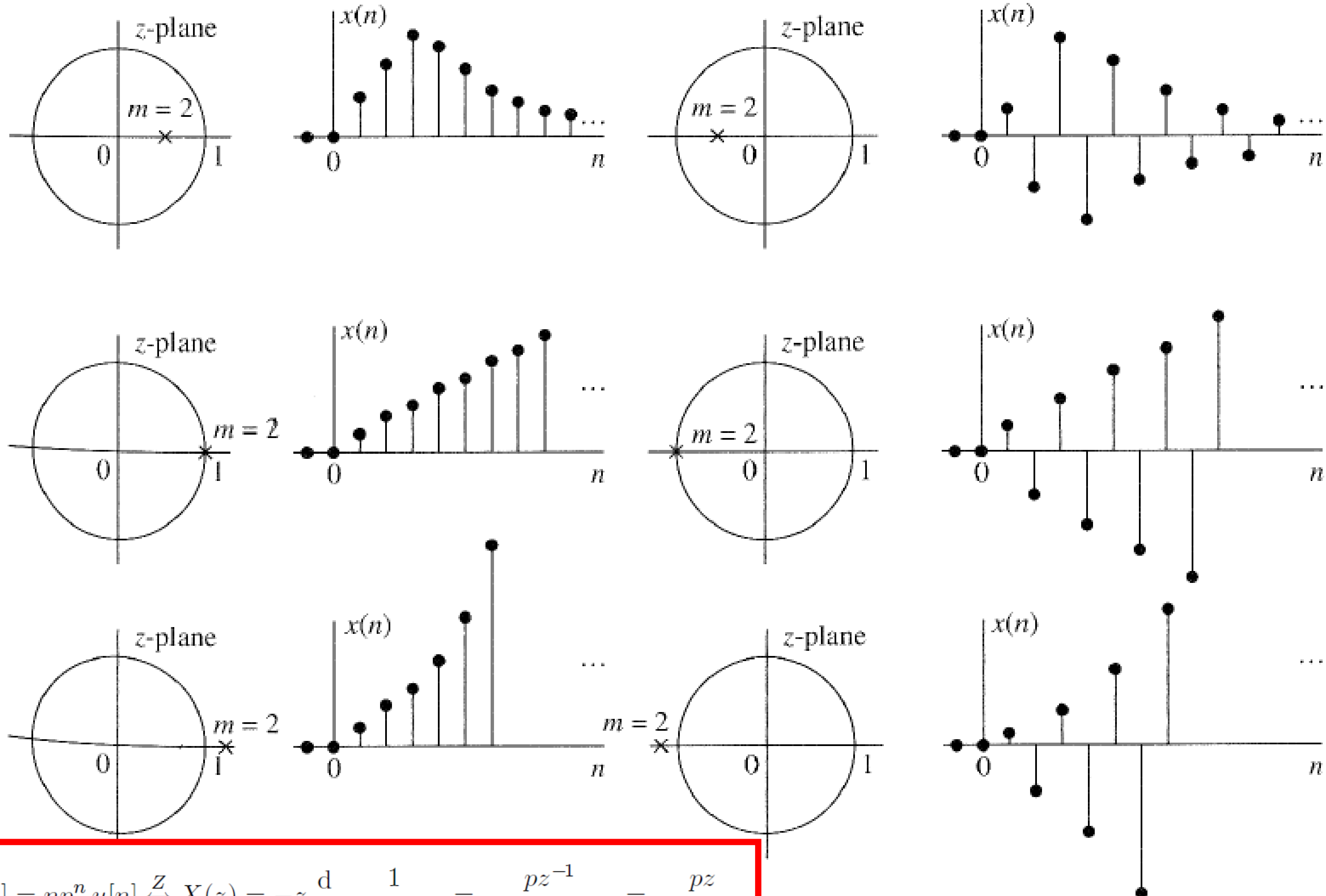
Rational **Z-Transform**

Pole Location and Time domain Behavior of real Causal Signals

Double Real pole:

$$x[n] = np^n\, u[n] \overset{Z}{\leftrightarrow} X(z) = -z\frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{1-pz^{-1}} = \frac{pz^{-1}}{(1-pz^{-1})^2} = \frac{pz}{(z-p)^2}$$

Rational Z-Transform



$$x[n] = np^n\, u[n] \overset{Z}{\leftrightarrow} X(z) = -z\frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{1-pz^{-1}} = \frac{pz^{-1}}{(1-pz^{-1})^2} = \frac{pz}{(z-p)^2}$$

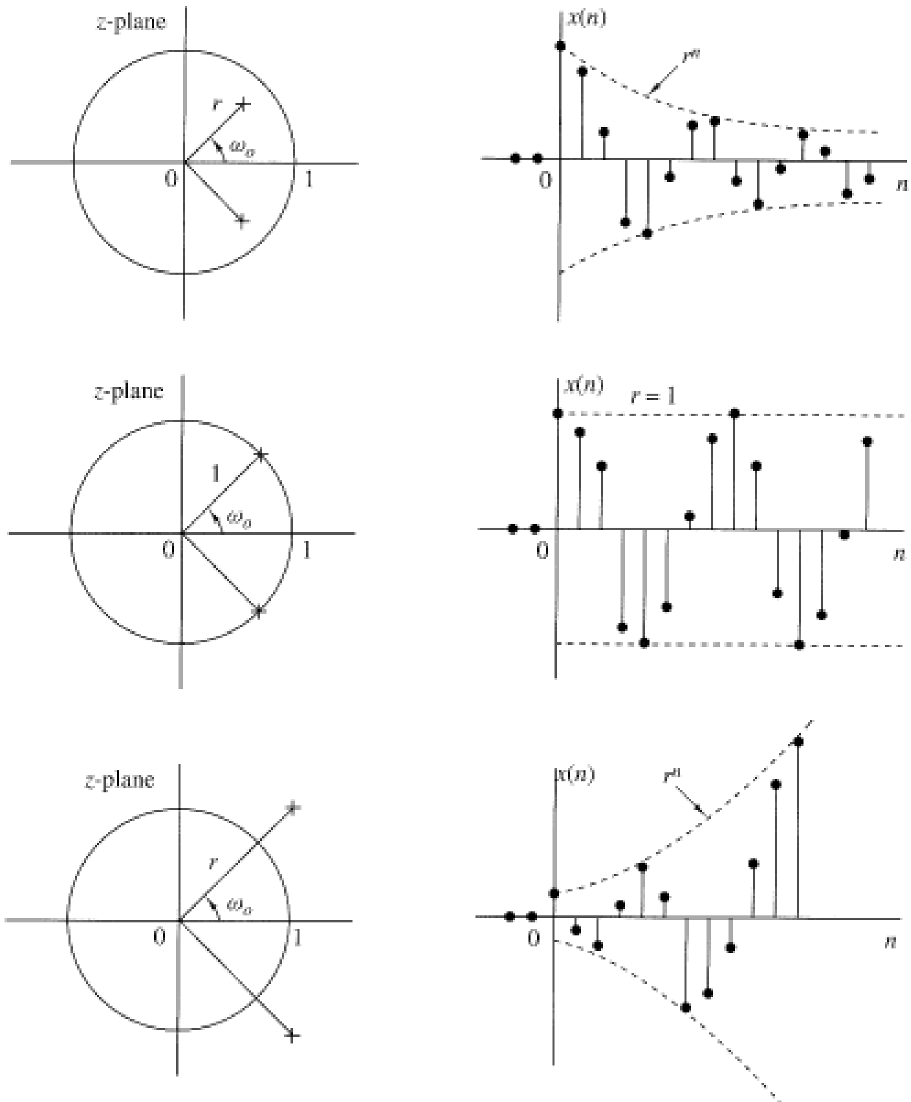
Rational **Z-Transform**

Pole Location and Time domain Behavior of real Causal Signals

Complex Conjugate poles:

$$x[n] = a^n \sin(\omega_0 n)\, u[n] \stackrel{Z}{\longleftrightarrow} X(z) = \frac{az \sin \omega_0}{(z-p)(z-p^*)}.$$

Rational **Z-Transform**



$$x[n] = a^n \sin(\omega_0 n)\, u[n] \overset{Z}{\longleftrightarrow} X(z) = \frac{az \sin \omega_0}{(z-p)(z-p^*)}.$$

Rational **Z-Transform**

Pole Location and Time domain Behavior of real Causal Signals

- **Causal Real signals with simple real poles or simple complex conjugate pairs of poles, which are inside or unit circle are always bounded in amplitude.**

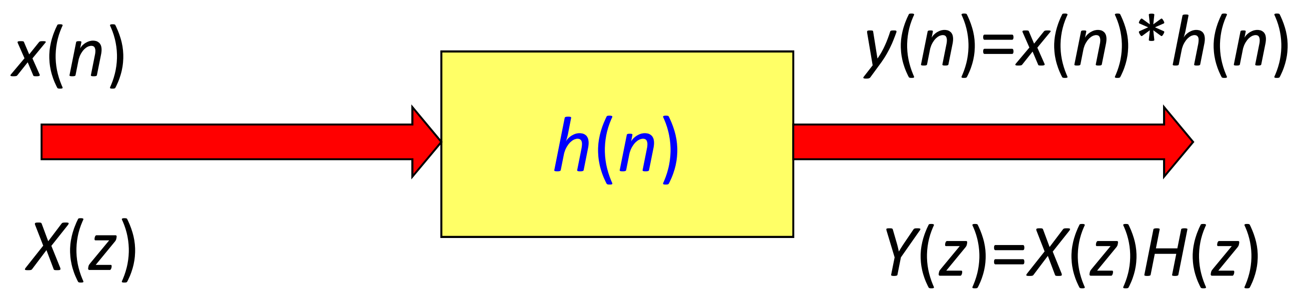
- **A signal with a pole/poles near the origin decays more rapidly than one associated with a pole/poles near but inside the unit circle.**

- **Time domain behavior strongly depends on the location of poles relative to unit circle.**

- **Zeros also effect time domain behavior but not so strongly.**

The System Function of an LTI system

The System Function



$$h(n) \overset{z}{\longleftrightarrow} H(z)$$

$$\text{time-domain} \overset{z}{\longleftrightarrow} z\text{-domain}$$

$$\text{impulse response} \overset{z}{\longleftrightarrow} \text{system function}$$

$$\boxed{H(z) = \frac{Y(z)}{X(z)}}$$

$$y(n) = x(n) * h(n) \overset{z}{\longleftrightarrow} Y(z) = X(z) \cdot H(z)$$

The System Function of an LTI system

The System Function of LCCDEs

$$
\begin{aligned}
y(n) &= -\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k) \\
\mathcal{Z}\{y(n)\} &= \mathcal{Z}\{-\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k)\} \\
\mathcal{Z}\{y(n)\} &= -\sum_{k=1}^{N} a_k \mathcal{Z}\{y(n-k)\} + \sum_{k=0}^{M} b_k \mathcal{Z}\{x(n-k)\} \\
Y(z) &= -\sum_{k=1}^{N} a_k z^{-k} Y(z) + \sum_{k=0}^{M} b_k z^{-k} X(z)
\end{aligned}
$$

The System Function of an LTI system

The System Function of LCCDEs

The system function for a LCCDE system is rational.

$$Y(z) + \sum_{k=1}^{N} a_k z^{-k} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$Y(z)\left[1 + \sum_{k=1}^{N} a_k z^{-k}\right] = X(z) \sum_{k=0}^{M} b_k z^{-k}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\left[1 + \sum_{k=1}^{N} a_k z^{-k}\right]}$$

$$\text{LCCDE} \longleftrightarrow \text{Rational System Function}$$

Many signals of practical interest have a rational z-Transform.

The System Function of an LTI system

System function of LCCDE: Outcome

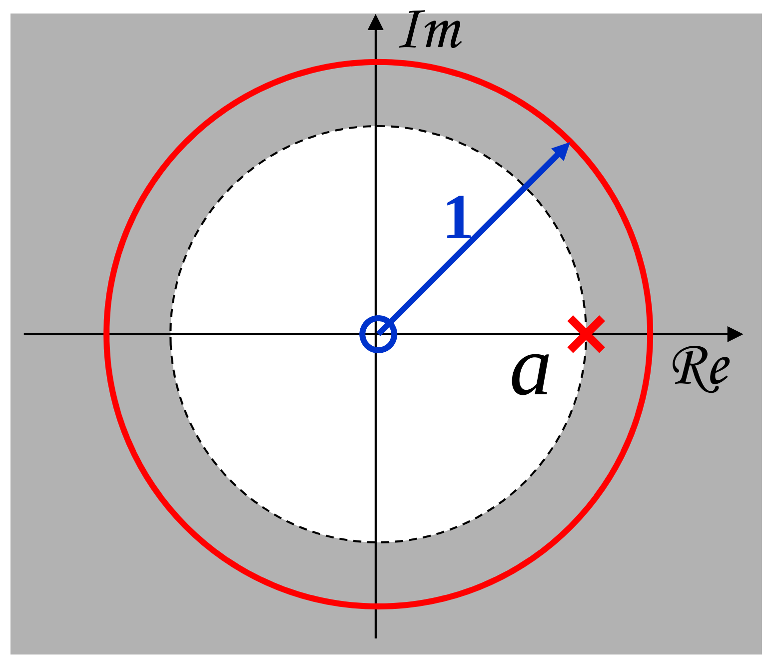
Convert between LCCDE and System function

Consider a causal and stable LTI system characterized by the following LCCDE:

$$y(n) = ay(n-1) + x(n)$$

$$\boxed{H(z) = \frac{1}{1-az^{-1}}}$$

$$h(n) = a^n u(n)$$



The System Function of an LTI system

Consider general form of system function i.e. :

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}},$$

The System Function of an LTI system

All Zero System:

$$\boxed{H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}},}$$

If $N = 0$ or equivalently $a_1 = \cdots = a_N = 0$, then the system function simplifies to

$$H(z) = \sum_{k=0}^{M} b_k z^{-k} = \frac{1}{z^M} \sum_{k=0}^{M} b_k z^{M-k} = \frac{\prod_{k=1}^{M}(z - z_k)}{z^M}$$

The M poles at $z = 0$ are called **trivial poles.**

Why are they called trivial poles?

One reason is that they correspond only to a time shift. The other is that if a system has a pole outside the unit circle, then certain bounded inputs will produce an unbounded output (unstable). But a pole at zero does not cause this unstable behavior, so its effect is in some sense trivial.

The System Function of an LTI system

All Zero System:

$$H(z) = \sum_{k=0}^{M} b_k z^{-k} = \frac{1}{z^M} \sum_{k=0}^{M} b_k z^{M-k} = \frac{\prod_{k=1}^{M} (z - z_k)}{z^M}$$

Then there are M roots of the numerator polynomial that are nontrivial zeros. Thus this is called a **all-zero system**. The impulse response is FIR:

$$h[n] = \sum_{k=0}^{N} b_k \, \delta[n-k]$$

The System Function of an LTI system

All Pole System:

$$\boxed{H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}},}$$

If $M = 0$ or equivalently $b_1 = \cdots = b_M = 0$, then the system function reduces to

$$H(z) = \frac{b_0}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \frac{b_0 z^N}{\sum_{k=0}^{N} a_k z^{N-k}} = b_0 \frac{z^N}{\prod_{k=1}^{N}(z - p_k)}$$

This system function has *N* trivial zeros at z = 0 that are relatively unimportant, and the denominator polynomial has *N* roots that are the poles of *H(z)*. Thus this is called an all-pole system. The impulse response is IIR.

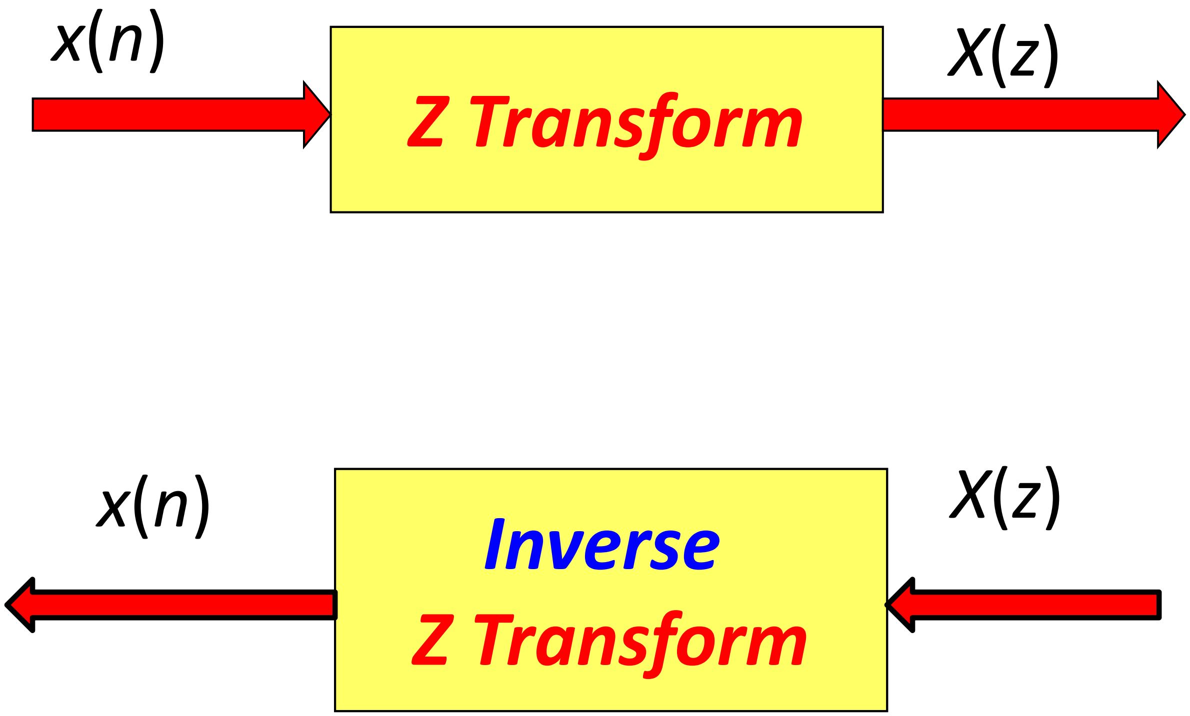
The System Function of an LTI system

Otherwise the impulse response is called a **pole-zero system**, and the impulse response is IIR.

Week 4: Z-Transform

- Direct Z-Transform and ROC
- Z-Transform Properties
- Rational Z-Transform
- **Inversion of Z-Transform**
- Analysis of LTI systems in Z-domain

Z-Transform Inversion



Z-Transform Inversion

- Direct Evaluation by contour integration

- Power Series Expansion

- **Partial Fraction expansion and table lookup**

Only this method will be covered !

Z-Transform Inversion

When *X(z)* is a rational function of *z* or *z*$^{-1}$ i.e. a ratio of two polynomials, the *inverse z-transform* can be obtained conveniently using partial fraction expansion

$$X(z) = \frac{N(z)}{D(z)} \Rightarrow X(z) = \alpha_1 X_1(z) + \alpha_2 X_2(z) + \ldots + \alpha_k X_k(z)$$

$X_1(z), X_2(z), \ldots, X_k(z) =$ Expressions for which we know their inverse

$\alpha_1, \alpha_2, \ldots, \alpha_k =$ Constants to e determined through partial fraction

$$x(n) = \alpha_1 x_1(n) + \alpha_2 x_2(n) + \ldots + \alpha_k x_k(n)$$

Z-Transform Inversion

Simple forms which will help in inverse z-transform computation in z^{-1} form

Type	$X(z)$	$x[n]$
polynomial in z	$\sum_k c_k z^{-k}$	$\sum_k c_k \, \delta[n-k]$
single real pole	$\dfrac{1}{1-pz^{-1}}$	$p^n \, u[n]$
double real pole	$\dfrac{pz^{-1}}{(1-pz^{-1})^2}$	$np^n \, u[n]$
double real pole	$\dfrac{1}{(1-pz^{-1})^2}$	$(n+1)p^n \, u[n]$
triple real pole	$\dfrac{1}{(1-pz^{-1})^3}$	$\dfrac{(n+2)(n+1)}{2} p^n \, u[n]$
complex conjugate pair	$\dfrac{az \sin \omega_0}{(z - a\, e^{\jmath\omega_0})(z - a\, e^{-\jmath\omega_0})}$	$a^n \sin(\omega_0 n) \, u[n]$
complex conjugate pair $p = \|p\| \, e^{\jmath\omega_0}$	$\dfrac{r}{1-pz^{-1}} + \dfrac{r^*}{1-p^* z^{-1}}$	$2\,\|r\|\,\|p\|^n \cos(\omega_0 n + \angle r) \, u[n]$

Z-Transform Inversion

$$X(z) = \frac{N(z)}{D(z)}$$

$$X(z) = \alpha_1 X_1(z) + \alpha_2 X_2(z) + \ldots + \alpha_k X_k(z)$$

$$x(n) = \alpha_1 x_1(n) + \alpha_2 x_2(n) + \ldots + \alpha_k x_k(n)$$

4 Steps

1. Decompose *X(z)* into proper form.
2. Find roots of denominator.
3. Partial fraction for distinct roots.
4. Inverse z-transform using table look up.

Z-Transform Inversion

1. Decompose *X(z)* into proper form:

$$X(z)=\frac{N(z)}{D(z)}=\frac{b_0+b_1z^{-1}+\ldots+b_Mz^{-M}}{a_0+a_1z^{-1}+\ldots+a_Nz^{-N}}=\frac{b_0+b_1z^{-1}+\ldots+b_Mz^{-M}}{1+a_1z^{-1}+\ldots+a_Nz^{-N}}$$

Proper Rational Function: $a_N=1$ and $M<N$

Improper Rational Function: $M\geq N$

We can always convert an improper rational function into sum of a polynomial and a proper rational function

We want to work with proper rational functions

$$\text{If } M\geq N\text{, then } \frac{P_M(z^{-1})}{P_N(z^{-1})}=P_{M-N}(z^{-1})+\frac{P_{N-1}(z^{-1})}{P_N(z^{-1})}.$$

Z-Transform Inversion

1. Decompose *X(z)* into proper form:

We can always convert an improper rational function into sum of a polynomial and a proper rational function

Example.

$$X(z) = \frac{1+z^{-2}}{1+2z^{-1}} = \frac{1}{2}z^{-1} + \left[\frac{1+z^{-2}}{1+2z^{-1}} - \frac{1}{2}z^{-1}\right] = \frac{1}{2}z^{-1} + \frac{1+z^{-2}-\frac{1}{2}z^{-1}\left(1+2z^{-1}\right)}{1+2z^{-1}}$$

$$= \frac{1}{2}z^{-1} + \frac{1-\frac{1}{2}z^{-1}}{1+2z^{-1}} = \frac{1}{2}z^{-1} - \frac{1}{4} + \left[\frac{1-\frac{1}{2}z^{-1}}{1+2z^{-1}} + \frac{1}{4}\right]$$

$$= -\frac{1}{4} + \frac{1}{2}z^{-1} + \frac{1-\frac{1}{2}z^{-1}+\frac{1}{4}\left(1+2z^{-1}\right)}{1+2z^{-1}} = -\frac{1}{4} + \frac{1}{2}z^{-1} + \frac{\frac{5}{4}}{1+2z^{-1}}$$

Z-Transform Inversion

2. Find roots i.e. poles of *X(z)* denominator:

$$X(z) = \frac{N(z)}{D(z)}$$

Suppose: $D(z) = c_1 + c_2 z^{-1} + c_3 z^{-2} = z^{-2}\left(c_1 z^2 + c_2 z + c_3\right)$

z^{-2} will go in numerator

So, it is same whether you factorize $c_1 + c_2 z^{-1} + c_3 z^{-2}$ OR $c_1 z^2 + c_2 z + c_3$

Factorization produces 2 roots i.e. poles of X(z) in this case: p_1 and p_2

Once you have p_1 and p_2 you can write A(z) into factorized form.

Z-Transform Inversion

$$D(z) = c_1 + c_2 z^{-1} + c_3 z^{-2} \quad = \quad z^{-2}\left(c_1 z^2 + c_2 z + c_3\right)$$

$$\downarrow \qquad\qquad\qquad\qquad \downarrow$$

$$D(z) = \left(1 - p_1 z^{-1}\right)\left(1 - p_2 z^{-1}\right) \quad = \quad z^{-2}\left(z - p_1\right)\left(z - p_2\right)$$

How to do factorization?

$$\left(p_1, p_2\right) = -\frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

After factorization you can write *X(z)* as sum of simplified functions either in *z* or z^{-1}

Z-Transform Inversion

3. Partial Fraction Expansion of *X(z)*:

Depends on type of poles

- **Single Order Poles**
- **Multiple Order poles**
- **Complex Conjugate Poles**

Z-Transform Inversion

3. Partial Fraction Expansion of *X(z)*:

Single order poles:

$$X(z) = \frac{r_1}{1 - p_1 z^{-1}} + \cdots + \frac{r_N}{1 - p_N z^{-1}}$$

$$\boxed{r_k = \left(1 - p_k z^{-1}\right) X(z)\Big|_{z=p_k}}$$

One can choose either *z* or *z*$^{-1}$ for partial fraction expansion. Book chooses *z*. BUT be careful in MATLAB you will use *residuez* command which chooses *z*$^{-1}$. Of course both will produce same *x[n]*.

$$\frac{X(z)}{z} = \frac{A_1}{z - p_1} + \frac{A_2}{z - p_2} + \cdots + \frac{A_N}{z - p_N}$$

$$\boxed{A_k = \frac{(z - p_k)X(z)}{z}\Big|_{z=p_k}}$$

Z-Transform Inversion

4. Inverse using table look up:

Single order poles:

$$X(z) = A_1 \frac{1}{1 - p_1 z^{-1}} + A_2 \frac{1}{1 - p_2 z^{-1}} + \cdots + A_N \frac{1}{1 - p_N z^{-1}}$$

$$Z^{-1}\left\{\frac{1}{1 - p_k z^{-1}}\right\} = \begin{cases} (p_k)^n u(n), & \text{if ROC: } |z| > |p_k| \\ & \text{(causal signals)} \\ -(p_k)^n u(-n-1), & \text{if ROC: } |z| < |p_k| \\ & \text{(anticausal signals)} \end{cases}$$

Z-Transform Inversion

- Single Order Poles
- Multiple Order poles
- Complex Conjugate Poles

Z-Transform Inversion

Single order poles:

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

$$X(z) = \frac{1}{(1 - 1z^{-1})(1 - 0.5z^{-1})} = \frac{A_1}{1 - 1z^{-1}} + \frac{A_2}{1 - 0.5z^{-1}}$$

$$A_1 = (1 - 1z^{-1})X(z)\Big|_{z^{-1}=1} = \frac{1}{(1 - 0.5z^{-1})}\Big|_{z^{-1}=1} = \frac{1}{1 - 0.5} = 2$$

$$A_2 = (1 - 0.5z^{-1})X(z)\Big|_{z^{-1}=2} = \frac{1}{(1 - z^{-1})}\Big|_{z^{-1}=2} = \frac{1}{1 - 2} = -1$$

$$X(z) = \frac{2}{1 - z^{-1}} - \frac{1}{1 - 0.5z^{-1}}$$

Assuming causal sequence

$$x(n) = \left[2 - (0.5)^n\right]u(n)$$

Z-Transform Inversion

Multiple order poles:

If X(z) has repeated poles. This time we use a different expansion.
Let's say D(z) contains a repeated poles $(1-pz^{-1})^r$. Then

$$X(z) = \frac{N(z)}{(1-pz^{-1})^r D_1(z)}$$
$$= \frac{A_1}{(1-pz^{-1})} + \frac{A_2 z^{-1}}{(1-pz^{-1})^2} + \ldots + \frac{A_{r-1} z^{-r+2}}{(1-pz^{-1})^{r-1}} + \frac{A_r z^{-r+1}}{(1-pz^{-1})^r} + \frac{N_1(z)}{D_1(z)}$$

To find $A_1, A_2, \ldots, A_r$

$$A_1 = \frac{1}{(r-1)!} \frac{d^{r-1}}{dz^{r-1}} \left[\frac{z^{r-1} N(z)}{D_1(z)} \right] \Bigg|_{z^{-1}=\frac{1}{p}}$$

$$A_2 = \frac{d}{dz} \left[\frac{z^{r-1} N(z)}{D_1(z)} \right] \Bigg|_{z^{-1}=\frac{1}{p}}$$

$$A_r = \frac{z^{r-1} N(z)}{D_1(z)} \Bigg|_{z^{-1}=\frac{1}{p}}$$

Z-Transform Inversion

Multiple order poles:

$$X(z)=\frac{1}{(1+z^{-1})(1-z^{-1})^2}$$

$$X(z)=\frac{1}{(1+z^{-1})(1-z^{-1})^2}=\frac{A_1}{(1+z^{-1})}+\frac{A_2}{(1-z^{-1})}+\frac{A_3 z^{-1}}{(1-z^{-1})^2}$$

$$A_1=(1+z^{-1})X(z)\Big|_{z^{-1}=-1}=\frac{1}{(1-z^{-1})^2}\Big|_{z^{-1}=-1}=\frac{1}{4}$$

$$A_3=(1-z^{-1})^2 zX(z)\Big|_{z^{-1}=1}=\frac{z}{(1+z^{-1})}\Big|_{z^{-1}=1}=\frac{1}{2}$$

$$A_2=\frac{d}{dz}\left[(1-z^{-1})^2 zX(z)\right]_{z^{-1}=1}$$

$$=\frac{d}{dz}\left[\frac{z}{(1+z^{-1})}\right]_{z^{-1}=1}=\frac{(1+z^{-1})-z(-z^{-2})}{(1+z^{-1})^2}\Big|_{z^{-1}=1}=\frac{2+1}{2^2}=\frac{3}{4}$$

Assuming causal sequence

$$X(z)=\frac{1}{4}\frac{1}{(1+z^{-1})}+\frac{3}{4}\frac{1}{(1-z^{-1})}+\frac{1}{2}\frac{z^{-1}}{(1-z^{-1})^2}$$

$$x(n)=\left[\frac{1}{4}(-1)^n+\frac{3}{4}+\frac{1}{2}n\right]u(n)$$

Z-Transform Inversion

Complex Conjugate Poles:

$$X(z) = \frac{1+z^{-1}}{1-z^{-1}+0.5z^{-2}}$$

$$\frac{X(z)}{z} = \frac{1+z^{-1}}{(1-p_1z^{-1})(1-p_2z^{-1})} = \frac{1+z^{-1}}{\left[1-\left(\frac{1}{2}+j\frac{1}{2}\right)z^{-1}\right]\left[1-\left(\frac{1}{2}-j\frac{1}{2}\right)z^{-1}\right]}$$

$$= \frac{A_1}{\left[1-\left(\frac{1}{2}+j\frac{1}{2}\right)z^{-1}\right]} + \frac{A_2}{\left[1-\left(\frac{1}{2}-j\frac{1}{2}\right)z^{-1}\right]}$$

Z-Transform Inversion

$$A_1 = (1 - p_1 z^{-1})X(z)\Big|_{z=p_1} = \left.\frac{1+z^{-1}}{\left[1-\left(\frac{1}{2}-j\frac{1}{2}\right)z^{-1}\right]}\right|_{z^{-1}=\frac{1}{\frac{1}{2}+j\frac{1}{2}}} = \frac{1+\dfrac{1}{\frac{1}{2}+j\frac{1}{2}}}{\left[1-\left(\frac{1}{2}-j\frac{1}{2}\right)\dfrac{1}{\frac{1}{2}+j\frac{1}{2}}\right]} = \frac{1}{2} - j\frac{3}{2}$$

$$A_2 = (1 - p_2 z^{-1})X(z)\Big|_{z=p_2} = \left.\frac{1+z^{-1}}{\left[1-\left(\frac{1}{2}+j\frac{1}{2}\right)z^{-1}\right]}\right|_{z^{-1}=\frac{1}{\frac{1}{2}-j\frac{1}{2}}} = \frac{1+\dfrac{1}{\frac{1}{2}-j\frac{1}{2}}}{\left[1-\left(\frac{1}{2}+j\frac{1}{2}\right)\dfrac{1}{\frac{1}{2}-j\frac{1}{2}}\right]} = \frac{1}{2} + j\frac{3}{2}$$

Z-Transform Inversion

$$X(z) = \frac{1+z^{-1}}{1-z^{-1}+0.5z^{-2}} = \frac{\frac{1}{2}-j\frac{3}{2}}{\left[1-\left(\frac{1}{2}+j\frac{1}{2}\right)z^{-1}\right]} + \frac{\frac{1}{2}+j\frac{3}{2}}{\left[1-\left(\frac{1}{2}-j\frac{1}{2}\right)z^{-1}\right]}$$

$$A_1 = \frac{1}{2} - j\frac{3}{2} = \frac{\sqrt{10}}{2}e^{-j71.565}$$

$$A_2 = \frac{1}{2} + j\frac{3}{2} = \frac{\sqrt{10}}{2}e^{+j71.565}$$

$$A_1 = A_2^*$$

$$p_1 = \frac{1}{2} + j\frac{1}{2} = \frac{1}{\sqrt{2}}e^{+j\pi/4}$$

$$p_1 = p_2^*$$

Z-Transform Inversion

$$X(z)=\frac{1+z^{-1}}{1-z^{-1}+0.5z^{-2}}=\frac{\frac{\sqrt{10}}{2}e^{-j71.565}}{\left[1-\frac{1}{\sqrt{2}}e^{j\pi/4}z^{-1}\right]}+\frac{\frac{\sqrt{10}}{2}e^{j71.565}}{\left[1-\frac{1}{\sqrt{2}}e^{-j\pi/4}z^{-1}\right]}$$

$$x(n)=\frac{\sqrt{10}}{2}e^{-j71.565}\left(\frac{1}{\sqrt{2}}e^{j\pi/4}\right)^{n}u(n)+\frac{\sqrt{10}}{2}e^{j71.565}\left(\frac{1}{\sqrt{2}}e^{-j\pi/4}\right)^{n}u(n)$$

$$x(n)=\sqrt{10}\left(\frac{1}{\sqrt{2}}\right)^{n}\left(\frac{1}{2}e^{j\left(\frac{\pi n}{4}-71.565\right)}+\frac{1}{2}e^{-j\left(\frac{\pi n}{4}-71.565\right)}\right)u(n)$$

$$x(n)=\sqrt{10}\left(\frac{1}{\sqrt{2}}\right)^{n}\cos\left(\frac{\pi n}{4}-71.565\right)u(n)$$

Assuming causal sequence

Z-Transform Inversion

Example:

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

ROC not specified

After Partial Fraction:

$$X(z) = \frac{2}{1 - z^{-1}} - \frac{1}{1 - 0.5z^{-1}}$$

3 choices in time domain

(a) ROC: $|z| > 1$

(b) ROC: $|z| < 0.5$

(c) ROC: $0.5 < |z| < 1$

Z-Transform Inversion

Be careful what is given and what is asked !

- **Determine the right sided sequence whose X(z) is given below**
- **Determine the left sided sequence whose X(z) is given below**
- **Determine the causal signal corresponding to the following X(z)**
- **Determine the anti-causal sequence whose z-transform is given below**
- **Determine the two sided sequence whose z-transform is given below**
- **Determine all possible signals in time domain corresponding to the following X(z)**
- **Determine x[n] corresponding to following pole-zero diagram**
- **etc.**

Z-Transform Inversion

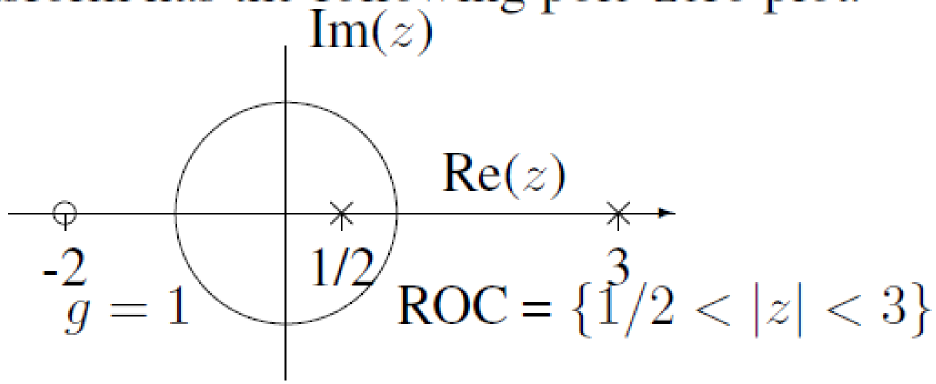
Example: **All possible signals in time domain**

$$Y(z) = \frac{1 - z^{-1}}{(1 - \frac{1}{2} z^{-1})(1 - \frac{1}{3} z^{-1})}$$

ROC	Signal
$\frac{1}{2} < \lvert z \rvert$	$-3(\frac{1}{2})^n u(n) + 4(\frac{1}{3})^n u(n)$
$\frac{1}{3} < \lvert z \rvert < \frac{1}{2}$	$3(\frac{1}{2})^n u(-n-1) + 4(\frac{1}{3})^n u(n)$
$\lvert z \rvert < \frac{1}{3}$	$3(\frac{1}{2})^n u(-n-1) - 4(\frac{1}{3})^n u(-n-1)$

Z-Transform Inversion

Find the signal $x[n]$ whose z-transform has the following pole-zero plot.



$$X(z) = \frac{z+2}{(z-3)(z-1/2)} = \frac{z^{-1}+2z^{-2}}{(1-3z^{-1})(1-\frac{1}{2}z^{-1})}$$

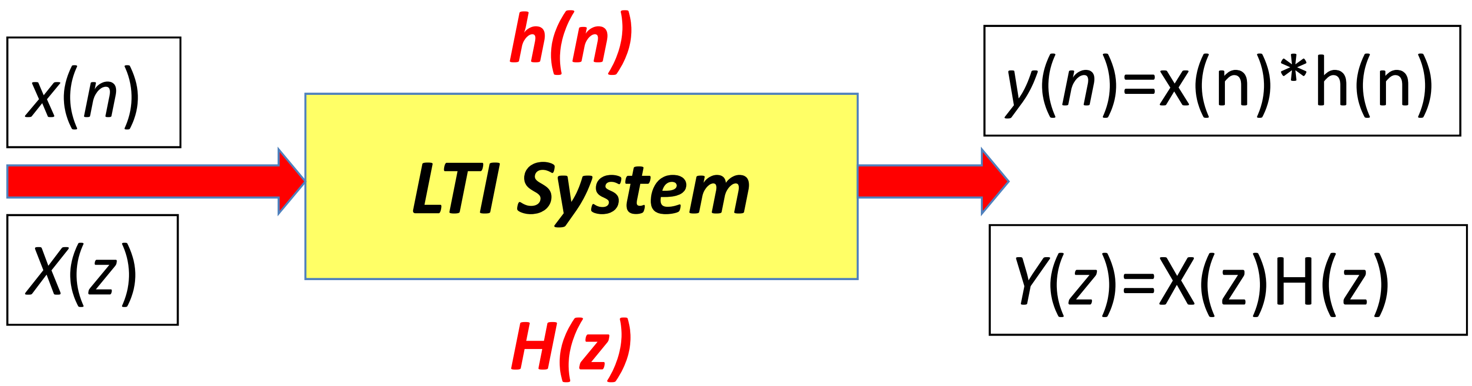
Verify at home !

$$x[n] = \frac{4}{3}\,\delta[n] - \frac{2}{3}3^n\,u[-n-1] - 2\left(\frac{1}{2}\right)^n u[n]$$

Week 4: Z-Transform

- Direct Z-Transform and ROC
- Z-Transform Properties
- Rational Z-Transform
- Inversion of Z-Transform
- **Analysis of LTI systems in Z-domain**

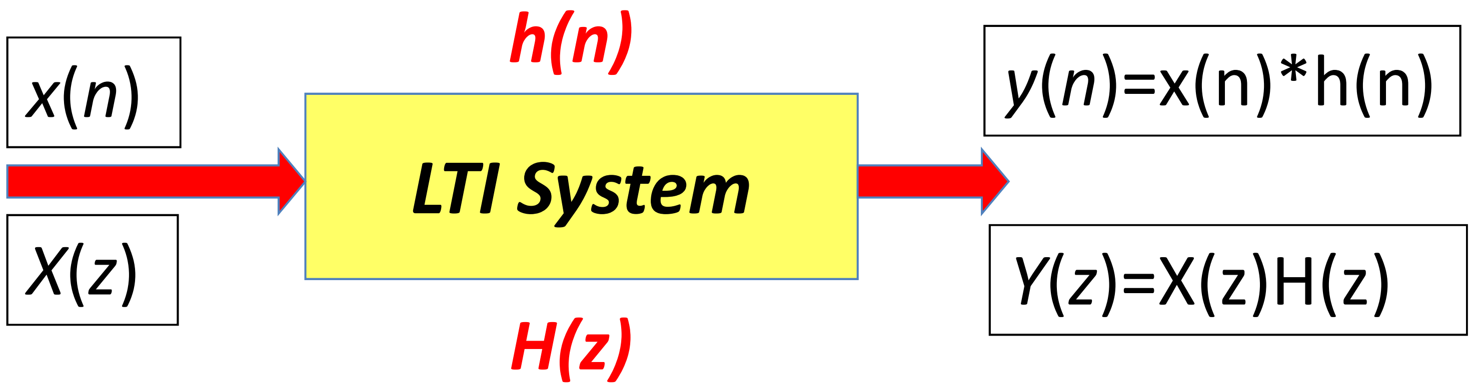
Analysis of LTI systems in Z-domain



Main Goal:

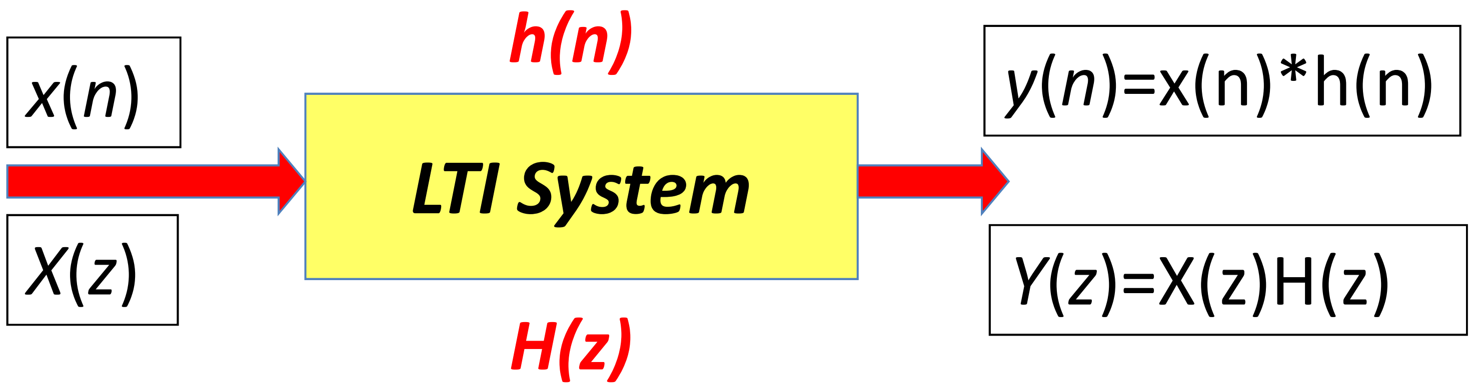
- Characterize Response to Inputs
- Characterize System Properties in Z-domain

Analysis of LTI systems in Z-domain



Characterize Response to Inputs

Analysis of LTI systems in Z-domain



H(z) for an LTI system is a rational function of type H(z)=B(z)/A(z). If X(z) is also a rational function of type X(z)=N(z)/D(z), then Y(z) is also a rational of the form Y(z)=B(z)N(z)/A(z)D(z)

Analysis of LTI systems in Z-domain

Natural Response vs. Forced response

Assume

- Poles of system $p_1, \ldots, p_N$ are unique
- Poles of input signal $q_1, \ldots, q_L$ are unique
- Poles of system and input signal are all different
- Zeros of system and input signal differ from all poles (so no pole-zero cancellation)
- Proper form
- Causal input sequence and causal LTI system

Then

$$X(z) = \sum_{k=1}^{L} \frac{\alpha_k}{1 - q_k z^{-1}} \xrightarrow{\mathcal{T}} Y(z) = \sum_{k=1}^{N} \frac{r_k}{1 - p_k z^{-1}} + \sum_{k=1}^{L} \frac{s_k}{1 - q_k z^{-1}}$$

so (assuming a causal system) the response is:

Due to input

$$y[n] = \underbrace{\sum_{k=1}^{N} r_k p_k^n \, u[n]}_{\text{natural}} + \underbrace{\sum_{k=1}^{L} s_k q_k^n \, u[n]}_{\text{forced}}.$$

Due to system itself

Analysis of LTI systems in Z-domain

Natural Response vs. Forced response

$$y[n] = \underbrace{\sum_{k=1}^{N} r_k p_k^n \, u[n]}_{\text{natural}} + \underbrace{\sum_{k=1}^{L} s_k q_k^n \, u[n]}_{\text{forced}} .$$

The output signal for a causal pole-zero system with input signal having rational z-transform is a weighted combination of geometric progression signals.

Analysis of LTI systems in Z-domain

Transient response

System Natural Response: $y_{\mathrm{nr}}[n] = \sum_{k=1}^{N} r_k p_k^n \, u[n]$

- If all the poles have magnitude less than unity then this response decays to zero as $n \to \infty$
- In such cases, we also natural response as transient response of the system.
- Smaller magnitude poles lead to faster signal decay. So, the closer the pole to the unit circle, the longer the transient response.

Analysis of LTI systems in Z-domain

Steady State response

System Forced Response: $y_{\text{fr}}[n] = \sum_{k=1}^{L} s_k q_k^n \, u[n]$

- If all of the input signal poles are within the unit circle, then the forced response will decay towards zero as $n \to \infty$
- If the input signal has a pole *on the unit circle* then there is a persistent sinusoidal component of the input signal. The forced response to such a sinusoid is also a persistent sinusoid.
- In this case, the forced response is also called the **steady-state** response.

Analysis of LTI systems in Z-domain

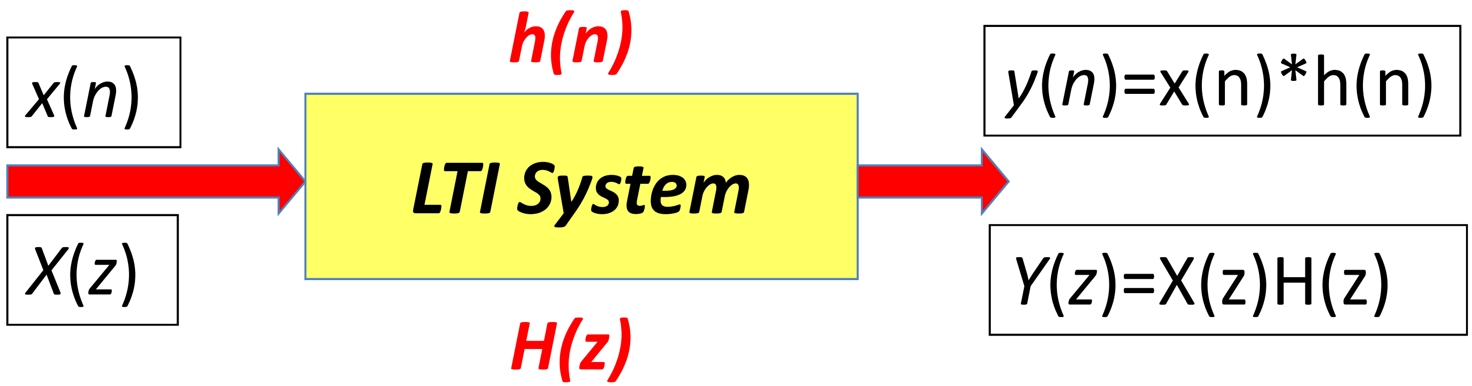
Example: Find Transient and steady state response of the following relaxed system for x[n]: $y[n] = 0.5y[n-1] + x[n]$ $x[n] = (-1)^n u(n)$

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \qquad X(z) = \frac{1}{1 + z^{-1}}$$

$$Y(z) = H(z)\,X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 + z^{-1})} = \frac{1}{1 + \frac{1}{2}z^{-1} - \frac{1}{2}z^{-2}} = \frac{1/3}{1 - \frac{1}{2}z^{-1}} + \frac{2/3}{1 + z^{-1}}$$

$$y[n] = \underbrace{\frac{1}{3}\left(\frac{1}{2}\right)^n u[n]}_{\text{natural / transient}} + \underbrace{\frac{2}{3}(-1)^n u[n]}_{\text{forced / steady state}}$$

Analysis of LTI systems in Z-domain

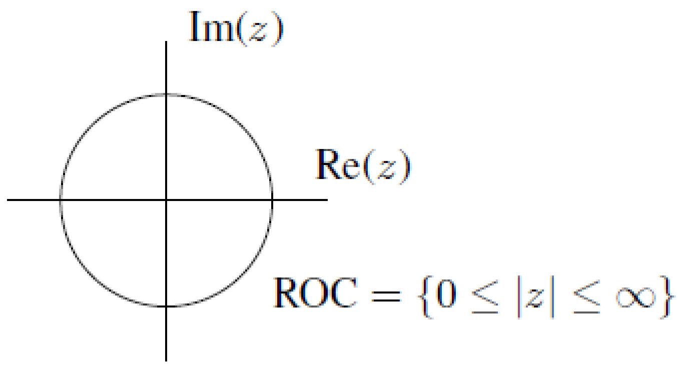


Characterize System Properties in Z-domain

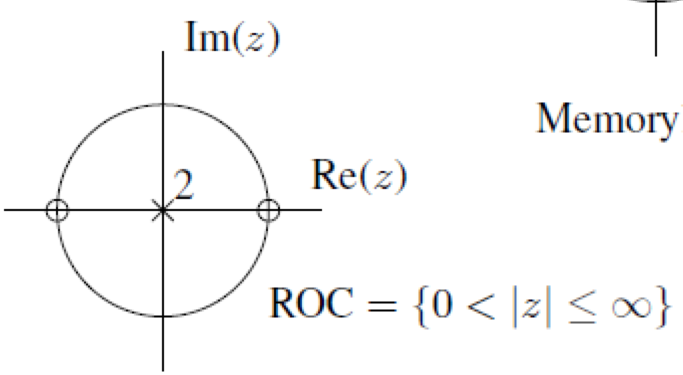
Analysis of LTI systems in Z-domain

Memory

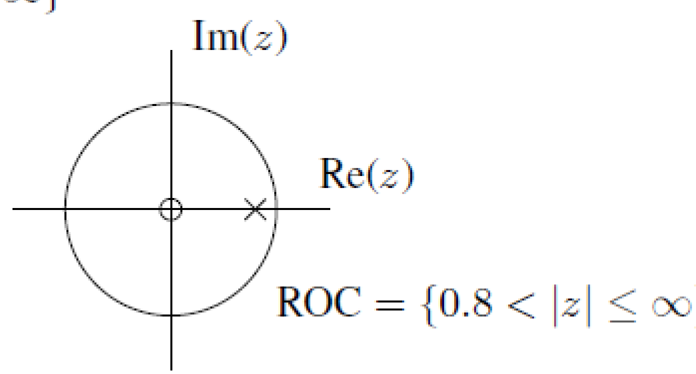
An LTI system is memoryless iff $h[n] = b_0\delta[n]$. So ***H(z) = b***$\boldsymbol{b_0}$. So H(z) has no poles or zeros, and ROC = Entire Z-Plane. In terms of dynamic systems, recall that previously we noted that FIR systems are all zero systems (poles at origin only).



Memoryless



FIR



IIR

Analysis of LTI systems in Z-domain

Causality

LTI system is causal iff its impulse response $\boxed{h[n] \text{ is } 0 \text{ for } n < 0.}$

ROC is exterior of a circle ⟵ **Right sided signal**

Is "ROC exterior of a circle" enough for causality? NO !

Example. $h[n] = u[n+1] \overset{Z}{\leftrightarrow} \dfrac{z}{1 - z^{-1}} = \dfrac{z^2}{z - 1}$ for $\{1 < |z| < \infty\}$.
The ROC is a circle's exterior, and $h[n]$ is right-sided, but the system is *not* causal.

Analysis of LTI systems in Z-domain

Causality

For a causal system, the system function (assuming it exists) has a series expansion that involves only *non-positive* powers of *z* :

$$H(z) = \sum_{n=0}^{\infty} h[n]\, z^{-n} = h[0] + h[1]\, z^{-1} + h[2]\, z^{-2} + \cdots$$

So the ROC of such an $H(z)$ will include $|z| = \infty$

An LTI system with impulse response *h[n]* is **causal** iff the ROC of the system function is the exterior of a circle of radius $r < \infty$ *including* $z = \infty$ i.e. ROC$=\{r < z \leq \infty\}$

Analysis of LTI systems in Z-domain

Causality

Is the LTI system with system function $H(z) = z^2 - z^{-1}$ causal?

ROC= Entire z plane except $z = 0$ and $z = \infty$

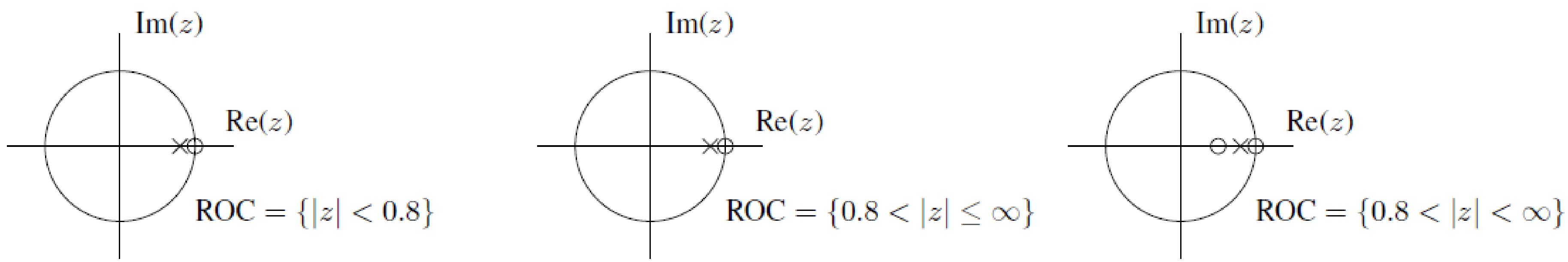
Since, $z = \infty$ not included, so, non-causal

Which is also clear if you find $h[n]$ which is $h[n] = \delta[n+2]$

Analysis of LTI systems in Z-domain

Causality

Example. Which of the following pole-zero plots correspond to causal systems?



Only the middle one. For the right one $H(z) = g\frac{(z-1)(z-1/2)}{z-0.8}$ which is infinite at $z = \infty$. It is noncausal.

A given pole-zero plot for a rational system function corresponds to a causal LTI system iff there are at least as many (finite) poles as (finite) zeros and the ROC is the exterior of the circle intersecting the outermost pole.

Analysis of LTI systems in Z-domain

Stability

LTI system is BIBO stable iff $\boxed{\sum_{n=-\infty}^{\infty} |h[n]| < \infty.}$

An LTI system is BIBO stable iff the ROC of its system function includes the unit circle.

Example. $y[n] = -y[n-1] + x[n] \Longrightarrow H(z) = \frac{1}{1+z^{-1}} = \frac{z}{z+1}$ which has a pole at $z = -1$ so this system is unstable

Analysis of LTI systems in Z-domain

Stability

- **In general causality and stability are unrelated properties.**
- **However, for a causal system we can narrow the condition for stability. For a causal system, the ROC is the exterior of a circle. For it to be stable as well, the ROC must include the unit circle, so the radius r for the ROC must be less than 1. There cannot be any poles in the ROC, so all the poles must be inside the circle of radius r = 1, which are thus inside the unit circle.**

A causal LTI system is BIBO stable iff all of the poles of its system function are *inside* the unit circle.

Example. Accumulator: $y[n] = y[n-1] + x[n]$ has $H(z) = \frac{1}{1-z^{-1}}$. Stable? No: causal but pole at $z = 1$ so unstable.

Analysis of LTI systems in Z-domain

Stability

Can poles of system function lie on the unit circle and still have the system be stable? No.

Example.

$$h[n] = u[n] \qquad H(z) = 1/(1 - z^{-1})$$

Now consider the input $x[n] = u[n]$ ← a bounded input

$$y[n] = (n+1)\,u[n]$$

So the output is not bounded

This can happen anywhere on the unit circle. Therefore for a causal system to be stable, all the poles of its system function must lie strictly inside the unit circle.

End